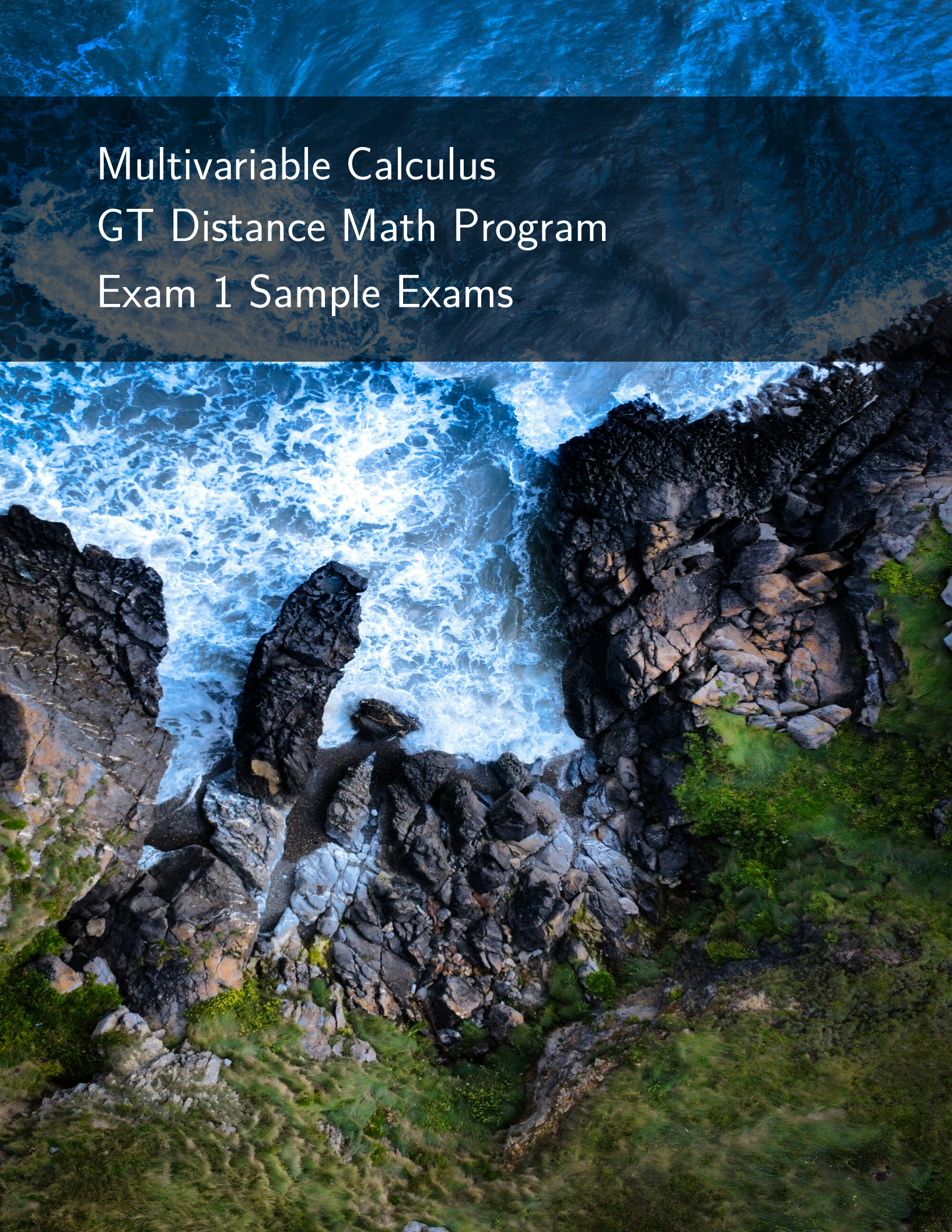


An aerial photograph of a rugged coastline. The top half of the image shows a dark blue, textured sea surface. Below this, a dark grey semi-transparent banner contains white text. The bottom half of the image shows a rocky shore with dark, jagged rocks and patches of green moss or algae. White, frothy waves are crashing against the rocks, creating a stark contrast with the dark water and rocks.

Multivariable Calculus GT Distance Math Program Exam 1 Sample Exams

Practice is the bridge between your abilities and success.

- Steve Nash

About This Document

The document was created using $\text{\LaTeX}$ and was last compiled on December 23, 2025.

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1 Recommended Study Strategies

1.1 General Advice on How to Prepare for Exams

- **Practice.** Work through many exercises. To understand linear algebra means being able to solve problems with linear algebra. Like any other skill, this needs to be developed through practice. Practice and persistence got you into this class, practice and persistence will get you through it.
- **Do More to Study Than Just the Sample Exams.** Studio worksheets, practice quizzes on Canvas, engaging in discussions on our forums, and old exams on Canvas are also great ways to study. Do not assume that any single resource will span the subspace that contains all concepts you need to know. Many linear algebra students for example look for additional exercises on true/false problems beyond what is in the sample exams.
- **Schedule Your Time.** Setting aside time with a digital or physical calendar helps you make sure you are making time for the learning you need to reach your goals.
- **Pace Yourself.** Linear algebra will draw connections between several inter-related concepts. Realize that understanding how concepts are related to each other, and how to apply them, can take time to learn. Pacing yourself helps you ensure that you have time to see these connections and ask help as you are learning them.
- **Reflect on Your Study Strategies.** Your strategies may need to adapt as you take more advanced courses that have you develop new skills. Identifying the strategies that are most efficient in getting you to reach your goals is a part of the journey.
- **Use Your Mistakes as a Guide.** This course can seem difficult at times. Keep pushing forward. Make the most of the mistakes: they tell you where you need to focus your time.
- **Develop Community.** Ask questions and answer questions. Attend office hours. The benefits of discussing course concepts with others are numerous. Beyond helping you clear up any specific confusion that you may have, you may realize there is something you didn't realize you needed to know more about, or identify areas of the course you need to work on.
- **Minimize Distractions When Studying.** Make the most of the time you set aside for the class, whether it is a lecture, studio, or studying for an exam. Try to put away anything that might pull you away from what you were there to do.
- **Read the Textbook.** Reading the textbook deepens your understanding of the material by helping you understand how the many different theorems and definitions we encounter interconnect. And reading is a skill that you will need in more advanced courses that have fewer support structures (not all courses have studios and online homework).
- **Familiarize Yourself With Exam Procedures.** The syllabus has everything you need to know about exams, including procedures for make-up exams, re-grade requests, topics covered on each exam and more. All students will need to be familiar with exam procedures. You should refer to the syllabus throughout the semester.
- **Take Care of Yourself.** Please take the time to care for your health as best you can. Everyone comes to this course with a different set of circumstances and interests. But anything you can do to stay healthy and be at your best can help you reach your goals.

2 Exams

2.1 Year 1 MATH 2551 Exam 1 Sample A

Work done on scratch paper will not be graded. Please justify your reasoning on all questions unless indicated otherwise. Good luck on the exam!

A Few Helpful Formulas

$$d = \left| \mathbf{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|, \quad d = \frac{|\mathbf{PS} \times \mathbf{v}|}{|\mathbf{v}|}, \quad \mathbf{N} = \frac{d\mathbf{T}'(t)/dt}{|d\mathbf{T}/dt|}, \quad \kappa = \frac{|\mathbf{T}'(t)|}{|\mathbf{v}|} = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}, \quad \kappa = \frac{|f''(x)|}{[1 + (f'(x))^2]^{3/2}}$$

1. (0.5 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID: , the High School you attend: .
2. (7 points) Fill in the blanks. You do not need to justify your reasoning.
- (a) If P is the point $(2, 8, 4)$, then the distance between P and the xy -plane is and the distance between P and the y -axis is .
- (b) The parallelogram determined by vectors $\mathbf{a}$ and $\mathbf{b}$ has area $\sqrt{2}$. If $\mathbf{a} = \mathbf{b} + \mathbf{c}$, and $\mathbf{c}$ makes an angle of $\frac{\pi}{4}$ with $\mathbf{b}$, and $|\mathbf{c}| = 6$, then $|\mathbf{b}| =$.
- (c) The cosine of the angle between the planes $2x - 2y - z = 1$ and $x + 2y + 2z = 2$ is .
- (d) A particle is moving along the path given by $x(t) = -t^2$, $y(t) = t^3 - 6t$. The largest interval over which the motion is up and to the right is $I = (a, b)$, where a is and b is .
- (e) The position of a moving object is given by the curve $\mathbf{r}(t) = \langle 1 - t, t^2, 3 \rangle$ for $t \in \mathbb{R}$. The speed of the object is given by . A unit vector that gives the direction of motion at time t is $\mathbf{T} = \langle f(t), g(t), h(t) \rangle$, where $f =$, $g =$, $h =$.
- (f) The velocity of a particle at time t is given by $\mathbf{v}(t) = \langle 4t, \sin(t), 2e^{2t} \rangle$ and the position of the object when $t = 0$ is $\mathbf{r} = \langle 1, 1, 1 \rangle$, then $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, where $f(t) =$, $g(t) =$, $h(t) =$.
- (g) The unit tangent vector for a curve $\mathbf{r}(t)$ is $\mathbf{T}(t) = \langle 0, \cos t, \sin t \rangle$. The unit normal vector is $\mathbf{N} = \langle f(t), g(t), h(t) \rangle$, where $f(t) =$, $g(t) =$, $h(t) =$.
3. (2 points) Indicate **true** if the statement is true, indicate **false** otherwise. You do not need to justify your reasoning.

	true	false
a) $\vec{u} \times \vec{v} = \vec{v} \times \vec{u}$ for any vectors $\vec{u}, \vec{v} \in \mathbb{R}^3$.	☐	☐
b) If $\mathbf{r}(t) = \langle t^2, 0 \rangle$ for $t \geq 0$, and $\mathbf{r}(t) = \langle 0, t^2 \rangle$ when $t < 0$, then $\mathbf{r}(t)$ is smooth.	☐	☐

4. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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, the High School you attend: .
5. (5 points) Suppose R is the point $R(0, 1, 2)$, L is the line $\mathbf{r}(t) = \langle t, t, t \rangle$. Please show your work.
- (a) Construct the equation of the plane that passes through R and contains the line L .

(b) Calculate the distance between the point R and the line L .

6. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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7. (5 points) An object is moving along a path given by $\mathbf{r}(t)$ for $t \geq 0$, where $\mathbf{r}'(t) = \mathbf{v} = \langle 4, t, 2\sqrt{2t} \rangle$. Please show your work for the parts below. You can use your results for each part using results from previous parts of this question.

(a) Determine the speed, acceleration vector, and unit tangent vector for the curve.

(b) Determine the length of the curve, L , from the point where $t = 4$ to $t = 8$. Simplify your answer as much as possible.

(c) Suppose the object starts at the origin, so that $\mathbf{r}(0) = \mathbf{0}$. Determine the value of t that corresponds to the point on the curve that is a distance of 10 units away from the origin along the curve. In other words, determine the value of t so that the arc length between the origin and $\mathbf{r}(t)$ is 10.

	true	false
a) $(\vec{u} \times \vec{v}) \times \vec{w} = \vec{u} \times (\vec{v} \times \vec{w})$ for any vectors $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^3$.	☐	☐
b) Two lines in $\mathbb{R}^n$ that are parallel to a third line in $\mathbb{R}^3$ must be parallel.	☐	☐

4. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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, the High School you attend: .
5. (5 points) Consider the point $S(4, 1, 1)$ and the line L whose parametric equations are $x = 2 + t$, $y = 1 - 3t$, $z = t$. Please show your work for the following.
- (a) Calculate the distance between S and L .
- (b) Determine the point P where L intersects the plane $x + y + 4z = 1$. Your answer should be expressed as a point in $\mathbb{R}^3$.

6. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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, the High School you attend: .

7. (5 points)

An object is moving along a path given by $\mathbf{r}(t) = \langle t, \cos t, \sin t \rangle$ for $t \in \mathbb{R}$. Please show your work for the parts below. You can use your results for each part using results from previous parts of this question.

- (a) Determine the velocity, acceleration, and unit tangent vectors for the curve.

- (b) Determine the length of the curve, L , from the point where $t = 4$ to $t = 12$.

- (c) Determine the curvature of $\mathbf{r}(t)$ when $t = 0$.

4. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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, the High School you attend: .
5. (5 points) A particle is located at the point $P(1, 2, 1)$ at time $t = 0$. It travels in a straight line to the point $Q(5, 6, 8)$, has speed 18 at $t = 0$, and acceleration $\mathbf{a} = 4\mathbf{i} + 4\mathbf{j} + 7\mathbf{k}$. Determine the velocity of the particle, $\mathbf{v}(t)$. Please show your work.

6. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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7. (5 points)

An object is moving in the xy -plane along a path given by $\mathbf{r}(t) = \langle t, t^2 + 1 \rangle$ for $t \in \mathbb{R}$. Please show your work for the parts below. You can use your results for each part using results from previous parts of this question.

(a) Determine the velocity, acceleration vector, and unit tangent vector for the curve.

(b) Determine the curvature of $\mathbf{r}(t)$ when $t = 0$.

(c) Given that the principal normal vector when $t = 0$ is $\mathbf{N} = \mathbf{j}$, determine the equation of the osculating circle when $t = 0$.

4. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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, the High School you attend: .

5. (5 points) Consider the two planes P_1 and P_2 .

$$P_1 : 2x + 3y + 6z = 2$$

$$P_2 : 4x + 6y + 12z = 6$$

Please show your work for the following.

- (a) Calculate the distance between planes P_1 and P_2 .

- (b) Determine the parametric equations for the line of intersection between plane P_1 and the plane P_3 , which is given by $2x + 4y + 8z = 4$.

6. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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, the High School you attend: .

7. (5 points)

An object is moving in the xy -plane along a path given by $\mathbf{r}(t) = \langle t, ((t-4)^2+1) \rangle = \langle t, t^2-8t+17 \rangle$ for $t \geq 0$. Please show your work for the parts below. You can use your results for each part using results from previous parts of this question.

(a) Determine the velocity and acceleration vectors for the curve for any $t \geq 0$.

(b) Determine the curvature of $\mathbf{r}(t)$ when $t = 4$.

(c) Determine the unit tangent vector $\mathbf{T}$ when $t = 4$.

(d) Given that the principal normal vector when $t = 4$ is $\mathbf{N} = \mathbf{j}$, determine the equation of the osculating circle when $t = 4$.

2.5 Year 1 MATH 2551 Exam 1 Sample E

Work done on scratch paper will not be graded. Please justify your reasoning on all questions unless indicated otherwise. Good luck on the exam!

A Few Helpful Formulas

$$d = \left| \mathbf{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|, \quad d = \frac{|\mathbf{PS} \times \mathbf{v}|}{|\mathbf{v}|}, \quad \mathbf{N} = \frac{d\mathbf{T}'(t)/dt}{|d\mathbf{T}/dt|}, \quad \kappa = \frac{|\mathbf{T}'(t)|}{|\mathbf{v}|} = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}, \quad \kappa = \frac{|f''(x)|}{[1 + (f'(x))^2]^{3/2}}$$

1. (0.5 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID: 9 0 , the High School you attend: .

2. (7 points) Fill in the blanks. You do not need to justify your reasoning.

- (a) The cosine of the angle between the vectors $\langle 4, 0, 3 \rangle$ and $\langle 2, 1, 2 \rangle$ is .
- (b) The parallelogram determined by vectors $\mathbf{a}$ and $\mathbf{b}$ has area 24. If $\mathbf{a} = \mathbf{b} + \mathbf{c}$, and $\mathbf{c}$ makes an angle of $\frac{\pi}{2}$ with $\mathbf{b}$, and $|\mathbf{c}| = 2$, then $|\mathbf{b}| =$.
- (c) The plane that passes through $P(5, 0, 2)$ and contains the line $x = 1 + 4t$, $y = t$, $z = 2 + 2t$ is .
- (d) Identify the surface $y + z^2 = 1 - x^2$ by type (cone, ellipsoid, etc.). .
- (e) The equation of the tangent line to the curve $\mathbf{r}(t) = \langle t - 1, 3t, t^2 \rangle$ at the point where $t = 2$ has equation $\mathbf{L} = \langle f(t), g(t), h(t) \rangle$, where $f(t) =$, $g(t) =$, $h(t) =$.
- (f) If the velocity of an object at time t is $\mathbf{v}(t) = \langle 6t^2, \cos(t), 4e^{2t} \rangle$ and its position at $t = 0$ is $\mathbf{r} = \langle 1, 2, 1 \rangle$, then $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, where $f(t) =$, $g(t) =$, $h(t) =$.
- (g) A plane curve $\mathbf{r}(t)$ has curvature $\kappa = \frac{1}{4}$ at the point $\mathbf{r}(t_0) = 3\mathbf{i} + 5\mathbf{j}$, and the unit normal vector at t_0 is $\mathbf{N} = \mathbf{i} = \langle 1, 0 \rangle$. The equation of the osculating circle at $t = t_0$ is .

3. (2 points) Indicate **true** if the statement is true, indicate **false** otherwise. You do not need to justify your reasoning.

true false

- a) It is possible to have more than n nonzero vectors in $\mathbb{R}^n$ which are all mutually orthogonal to each other. ☐ ☐
- b) Two lines in $\mathbb{R}^3$ are either parallel or they intersect each other. ☐ ☐

4. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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, the High School you attend: .
5. (5 points) A particle is located at the point $P(1, 2, 5)$ at time $t = 0$. It travels in a straight line to the point $Q(2, 4, 7)$, has speed 12 at $t = 0$, and acceleration $\mathbf{a} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$. Determine the velocity of the particle, $\mathbf{v}(t)$. Please show your work.

6. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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7. (5 points)

An object is moving along a path given by $\mathbf{r}(t) = \langle t, \cos t, \sin t \rangle$ for $t \in \mathbb{R}$. Please show your work for the parts below. You can use your results for each part using results from previous parts of this question.

- (a) Determine the velocity, acceleration, and unit tangent vectors for the curve.

- (b) Determine the length of the curve, L , from the point where $t = 1$ to $t = 4$.

- (c) Determine the curvature of $\mathbf{r}(t)$ when $t = 0$.

2.6 Year 1 MATH 2551 Exam 1 Sample F

Work done on scratch paper will not be graded. Please justify your reasoning on all questions unless indicated otherwise. Good luck on the exam!

A Few Helpful Formulas

$$d = \left| \mathbf{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|, \quad d = \frac{|\mathbf{PS} \times \mathbf{v}|}{|\mathbf{v}|}, \quad \mathbf{N} = \frac{d\mathbf{T}'(t)/dt}{|d\mathbf{T}/dt|}, \quad \kappa = \frac{|\mathbf{T}'(t)|}{|\mathbf{v}|} = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}, \quad \kappa = \frac{|f''(x)|}{[1 + (f'(x))^2]^{3/2}}$$

- (0.5 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID: 90, the High School you attend: .
- (7 points) Fill in the blanks. You do not need to justify your reasoning.
 - The projection of the point $P(2, 3, 4)$ onto the yz -plane is the point $Q(c_1, c_2, c_3)$ where $c_1 = \text{}$, $c_2 = \text{}$, $c_3 = \text{}$.
 - The cosine of the angle between the vectors $\mathbf{a} = \langle 2, 1, 2 \rangle$ and $\mathbf{b} = \langle 1, 0, 2 \rangle$ is $\cos \theta = \text{}$.
 - The distance between the plane $x + 4y + 8z = 1$ and the point $S(28, 0, 0)$ is .
 - Identify the surface $-x^2 + y^2 + 4z^2 = 0$ by type (cone, paraboloid, etc.). .
 - The tangent line to the curve $\mathbf{r}(t) = \langle 4t - 1, t^3, t^2 \rangle$ at the point where $t = 2$ is $\mathbf{L} = \langle f(t), g(t), h(t) \rangle$, where $f(t) = \text{}$, $g(t) = \text{}$, $h(t) = \text{}$.
 - A projectile is launched from the ground with an initial speed of 500 m/sec at an angle of elevation of θ , where $\sin \theta = \frac{1}{2}$. If we use $g = 10 \text{ m/s}^2$ as the acceleration due to gravity, the object will return to the ground when $t = \text{}$ seconds.
 - The velocity vector for an object moving on the curve $\mathbf{r}(t)$ is $\mathbf{v}(t) = \langle 4t \cos t, 4t \sin t \rangle$ for $t > 0$. The speed of the object for $t > 0$ is $s(t) = |\mathbf{v}| = \text{}$. The unit tangent vector for $t > 0$ is $\mathbf{T} = \langle f(t), g(t) \rangle$, where $f(t) = \text{}$, $g(t) = \text{}$. The curvature is $\kappa = \text{}$.
- (2 points) Indicate **true** if the statement is true, indicate **false** otherwise. You do not need to justify your reasoning.

true false

- | | | |
|---|-----------------------|-----------------------|
| a) The surface defined by the equation $y = x^2$ in $\mathbb{R}^3$ is an example of a cylinder. | ☐ | ☐ |
| b) The trace obtained from the intersection of the plane $z = 1$ and the surface $(x/2)^2 + (y/3)^2 + (z/4)^2 = 4$ is a circle. | ☐ | ☐ |

4. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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, the High School you attend: .
5. (5 points) Consider the point $S(4, 1, 1)$ and the L_1 whose parametric equations are $x = 2 + t$, $y = 1 - t$, $z = t$. Please show your work for the following.

(a) Calculate the distance between the point and line L_1 .

(b) Construct the component equation of the plane that contains L_1 and the line L_2 whose parametric equations are $x = 2 + 2t$, $y = 1 + 4t$, $z = 3t$.

6. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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, the High School you attend: .

7. (5 points)

An object is moving along a path given by $\mathbf{r}(t)$ for $t \geq 0$, where $\mathbf{r}'(t) = \mathbf{v} = \langle t, 3, \sqrt{6t} \rangle$. Please show your work for the parts below. You can use your results for each part using results from previous parts of this question.

- (a) Determine the speed, acceleration vector, and unit tangent vector for the curve.
- (b) Determine the length of the curve, L , from the point where $t = 2$ to $t = 4$.
- (c) Suppose the object starts at the origin, so that $\mathbf{r}(0) = \mathbf{0}$. Determine the value of t that corresponds to the point on the curve that is a distance of 8 units away from the origin along the curve. In other words, determine the value of t so that the arc length between the origin and $\mathbf{r}(t)$ is 8.

- a) The surface defined by the equation $z^2 = x^2 + y^2$ is an example of a cylinder. ☐ ☐
- b) The surface $5x^2 + 8y^2 - z = 0$ is an elliptical paraboloid whose axis of symmetry is the z -axis. ☐ ☐

4. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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, the High School you attend: .
5. (5 points) Consider the point $S(4, 3, 1)$ and the L_1 whose parametric equations are $x = 2 + t$, $y = 1 - 2t$, $z = 2t$. Please show your work for the following.

(a) Calculate the distance between the point and line L_1 .

(b) Construct the component equation of the plane that contains L_1 and the line L_2 . The line L_2 has the parametric equations $x = 2 + 2t$, $y = 1$, $z = -3t$.

6. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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, the High School you attend: .

7. (5 points)

An object is moving along a path given by $\mathbf{r}(t)$ for $t \geq 0$, where $\mathbf{r}'(t) = \mathbf{v} = \langle t, 1, \sqrt{2t} \rangle$. Please show your work for the parts below. You can use your results for each part using results from previous parts of this question.

- (a) Determine the speed and the acceleration vector for the curve.
- (b) Determine the length of the curve, L , from the point where $t = 2$ to $t = 8$. Simplify your answer as much as possible.
- (c) Suppose the object starts at the origin, so that $\mathbf{r}(0) = \mathbf{0}$. Determine the value of t that corresponds to the point on the curve that is a distance of 6 units away from the origin along the curve. In other words, determine the value of t so that the arc length between the origin and $\mathbf{r}(t)$ is 6. Please show your work.

2.8 Year 1 MATH 2551 Exam 1 Sample H

Work done on scratch paper will not be graded. Please justify your reasoning on all questions unless indicated otherwise. Good luck on the exam!

A Few Helpful Formulas

$$d = \left| \mathbf{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|, \quad d = \frac{|\mathbf{PS} \times \mathbf{v}|}{|\mathbf{v}|}, \quad \mathbf{N} = \frac{d\mathbf{T}'(t)/dt}{|d\mathbf{T}/dt|}, \quad \kappa = \frac{|\mathbf{T}'(t)|}{|\mathbf{v}|} = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}, \quad \kappa = \frac{|f''(x)|}{[1 + (f'(x))^2]^{3/2}}$$

1. (0.5 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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, the High School you attend:

2. (7 points) Fill in the blanks. You do not need to justify your reasoning.

- (a) The cosine of the angle between the vectors $\langle 2, 3, -6 \rangle$ and $\langle 2, 2, 1 \rangle$ is $\cos \theta =$.
- (b) The volume of the parallelepiped determined by $\langle 1, 0, 1 \rangle, \langle 1, 1, 0 \rangle, \langle 0, 0, 2 \rangle$ is .
- (c) The plane that passes through $P(0, -2, 2)$ and contains the line $x = 1 + t, y = 2t, z = 2 - t$ is . The distance between P and the xz -plane is .
- (d) Identify the surface $-x^2 - y^2 + z = 0$ by type (cone, ellipsoid, etc.). .
- (e) The position of a moving object is given by the curve $\mathbf{r}(t) = \langle 1 - 2t, 6t^3 \rangle$ for $t \in \mathbb{R}$. The curve passes through the point $P(5, -48)$ when $t =$. The times when the direction of motion is perpendicular to $\mathbf{w} = \langle 1, 1 \rangle$, are $t =$ and $t =$.
- (f) If the velocity of an object at time t is $\mathbf{v}(t) = \langle 4t^3, \cos(t), 4e^{2t} \rangle$ and its position at $t = 0$ is $\mathbf{r} = \langle 3, 5, 14 \rangle$, then $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, where $f(t) =$, $g(t) =$, $h(t) =$.
- (g) A plane curve $\mathbf{r}(t)$ has curvature $\kappa = \frac{1}{4}$ at $\mathbf{r}(t_0) = \langle 5, 3 \rangle$, and the unit normal vector at $t = t_0$ is $\mathbf{N} = \mathbf{j} = \langle 0, 1 \rangle$. The equation of the osculating circle at $t = t_0$ is .

3. (2 points) Indicate **true** if the statement is true, indicate **false** otherwise. You do not need to justify your reasoning.

	true	false
a) The set of points equidistant from a coordinate plane and a point off of that plane in $\mathbb{R}^3$ forms an elliptic paraboloid.	☐	☐
b) The surface $-3x^2 + 5y^2 - z^2 = 1$ is a cone along the z -axis.	☐	☐

4. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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5. (5 points) Suppose R is the point $R(0, 1, 2)$, L is the line $\mathbf{r}(t) = \langle 3t, 2, 4t \rangle$. Please show your work.

(a) Construct the equation of the plane that passes through R and contains the line L .

(b) Calculate the distance between the point R and the line L .

6. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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7. (5 points)

Consider the curve $\mathbf{r}(t) = 6 \cos t \mathbf{i} + 6 \sin t \mathbf{j} + 8t \mathbf{k}$, $t \in \mathbb{R}$. Please show your work for the following.

- a) Determine the unit tangent vector for this curve.

- b) Determine the arc length parameter $s(t)$ along the curve from the point where $t_0 \in \mathbb{R}$ by evaluating the integral

$$s = s(t) = \int_{t_0}^t \left| \frac{d\mathbf{r}}{d\tau} \right| d\tau = \int_{t_0}^t |\mathbf{v}| d\tau$$

- c) The point $P_0(6, 0, 16\pi)$ is a point on the curve that corresponds to $t_0 = 2\pi$. Use your results from part (b) to determine the coordinates of any point on the curve that is a distance of 10π units **along the curve** from P_0 . Give your answers in terms of a point in $\mathbb{R}^3$.

4. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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5. (5 points)

Consider the point $S(6, 5, 2)$ and the L_1 whose parametric equations are $x = 2 + t$, $y = 1 - 2t$, $z = 2t$. Please show your work for the following.

- (a) Calculate the distance between the point and line L_1 .

- (b) Construct the component equation of the plane that contains L_1 and the line L_2 . The line L_2 has the parametric equations $x = 2 + 2t$, $y = 1$, $z = -3t$.

6. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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7. (5 points)

Consider the curve $\mathbf{r}(t) = 6 \cos t \mathbf{i} + 6 \sin t \mathbf{j} + 8t \mathbf{k}$, $t \in \mathbb{R}$. Please show your work for the following.

- a) Determine the unit tangent vector for this curve.

- b) Determine the arc length parameter $s(t)$ along the curve from the point where $t_0 \in \mathbb{R}$ by evaluating the integral

$$s = s(t) = \int_{t_0}^t \left| \frac{d\mathbf{r}}{d\tau} \right| d\tau = \int_{t_0}^t |\mathbf{v}| d\tau$$

- c) The point $P_0(6, 0, 16\pi)$ is a point on the curve that corresponds to $t_0 = 2\pi$. Use your results from part (b) to determine the coordinates of any point on the curve that is a distance of 10π units **along the curve** from P_0 . Give your answers in terms of a point in $\mathbb{R}^3$.

2.10 Year 1 MATH 2551 Exam 1 Sample J

Work done on scratch paper will not be graded. Please justify your reasoning on all questions unless indicated otherwise. Good luck on the exam!

A Few Helpful Formulas

$$d = \left| \mathbf{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|, \quad d = \frac{|\mathbf{PS} \times \mathbf{v}|}{|\mathbf{v}|}, \quad \mathbf{N} = \frac{d\mathbf{T}'(t)/dt}{|d\mathbf{T}/dt|}, \quad \kappa = \frac{|\mathbf{T}'(t)|}{|\mathbf{v}|} = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}, \quad \kappa = \frac{|f''(x)|}{[1 + (f'(x))^2]^{3/2}}$$

1. (0.5 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID: , the High School you attend: .
2. (7 points) Fill in the blanks. You do not need to justify your reasoning.

(a) The equation $x^2 + y^2 - 6y + z^2 + 4z = 12$ represents a sphere whose radius is $r = \boxed{}$. The center of the sphere is at the point $P(a, b, c)$ where $a = \boxed{}$, $b = \boxed{}$, $c = \boxed{}$.

(b) The parallelogram determined by vectors $\mathbf{a}$ and $\mathbf{b}$ has area 12. If $\mathbf{a} = \mathbf{b} + \mathbf{c}$, and $\mathbf{c}$ makes an angle of $\frac{\pi}{2}$ with $\mathbf{b}$, and $|\mathbf{c}| = 2$, then $|\mathbf{b}| = \boxed{}$.

(c) The plane that passes through $P(1, 3, 2)$ and contains the line $x = 1 + t$, $y = 2 + 3t$, $z = 2 + 2t$ is . The distance between the point P and the x -axis is .

(d) Identify the surface $x - 5y^2 - 2z^2 = 0$ by type (cone, ellipsoid, etc.). .

(e) The position of a moving object is given by the curve $\mathbf{r}(t) = \langle 1 - t, 4, 3t \rangle$ for $t \in \mathbb{R}$. The speed of the object is given by . A unit vector that gives the direction of motion at time t is $\mathbf{T} = \langle f(t), g(t), h(t) \rangle$, where $f = \boxed{}$, $g = \boxed{}$, $h = \boxed{}$.

(f) Consider the initial value problem (IVP): $\mathbf{r}' = 32t\mathbf{i} + 6t^2\mathbf{j} + 4e^{2t}\mathbf{k}$, $\mathbf{r}(0) = 3\mathbf{i} + 5\mathbf{j} + 8\mathbf{k}$. The solution to the IVP is $\mathbf{r} = \langle f(t), g(t), h(t) \rangle$, where $f(t) = \boxed{}$, $g(t) = \boxed{}$, and $h(t) = \boxed{}$.

(g) A plane curve $\mathbf{r}(t)$ has curvature $\kappa = \frac{1}{10}$ at $\mathbf{r}(t_0) = \langle 5, 0 \rangle$, and the unit normal vector at $t = t_0$ is $\mathbf{N} = \mathbf{j} = \langle 0, 1 \rangle$. The equation of the osculating circle at $t = t_0$ is .

3. (2 points) Indicate **true** if the statement is true, indicate **false** otherwise. You do not need to justify your reasoning.

true false

- a) If $\mathbf{r}(t)$ is a differentiable vector function of t and $\|\mathbf{r}\|$ is constant, then $\mathbf{r} \cdot \mathbf{r}' = 0$. ☐ ☐
- b) Two lines in $\mathbb{R}^3$ are that are parallel to a plane are parallel to each other. ☐ ☐

4. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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5. (5 points)

Suppose R is the point $R(0, 1, 2)$, L is the line $\mathbf{r}(t) = \langle 3t, 2, 4t \rangle$. Please show your work.

(a) Construct the equation of the plane that passes through R and contains the line L .

(b) Calculate the distance between the point R and the line L .

6. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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7. (5 points)

An object is moving along a path given by $\mathbf{r}(t)$ for $t \geq 0$, where $\mathbf{r}'(t) = \mathbf{v} = \langle t, 1, \sqrt{2t} \rangle$. Please show your work for the parts below. You can use your results for each part using results from previous parts of this question.

- (a) Determine the speed and the acceleration vector for the curve.
- (b) Determine the length of the curve, L , from the point where $t = 2$ to $t = 8$. Simplify your answer as much as possible.
- (c) Suppose the object starts at the origin, so that $\mathbf{r}(0) = \mathbf{0}$. Determine the value of t that corresponds to the point on the curve that is a distance of 6 units away from the origin along the curve. In other words, determine the value of t so that the arc length between the origin and $\mathbf{r}(t)$ is 6. Please show your work.

3 Solutions

Solutions

3.1 Year 1 MATH 2551 Exam 1 Sample A

Work done on scratch paper will not be graded. Please justify your reasoning on all questions unless indicated otherwise. Good luck on the exam!

A Few Helpful Formulas

$$d = \left| \mathbf{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|, \quad d = \frac{|\mathbf{PS} \times \mathbf{v}|}{|\mathbf{v}|}, \quad \mathbf{N} = \frac{d\mathbf{T}'(t)/dt}{|d\mathbf{T}/dt|}, \quad \kappa = \frac{|\mathbf{T}'(t)|}{|\mathbf{v}|} = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}, \quad \kappa = \frac{|f''(x)|}{[1 + (f'(x))^2]^{3/2}}$$

1. (0.5 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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2. (7 points) Fill in the blanks. You do not need to justify your reasoning.

- (a) If P is the point $(2, 8, 4)$, then the distance between P and the xy -plane is and the distance between P and the y -axis is .

Answer: the point is 4 units above the xy -plane, so the first distance is 4. Looking down the y -axis, the point is 2 units to the left of the y -axis and 8 units above it, so using a right-angle triangle and the Pythagorean theorem the point is $\sqrt{2^2 + 4^2} = \sqrt{20}$ units away for the y -axis.

- (b) The parallelogram determined by vectors $\mathbf{a}$ and $\mathbf{b}$ has area $\sqrt{2}$. If $\mathbf{a} = \mathbf{b} + \mathbf{c}$, and $\mathbf{c}$ makes an angle of $\frac{\pi}{4}$ with $\mathbf{b}$, and $|\mathbf{c}| = 6$, then $|\mathbf{b}| =$.

Answer: $1/3$

Solutions:

$$\begin{aligned}
 \sqrt{2} &= |\mathbf{a} \times \mathbf{b}| \\
 &= |(\mathbf{b} + \mathbf{c}) \times \mathbf{b}| \\
 &= |\mathbf{b} \times \mathbf{b} + \mathbf{c} \times \mathbf{b}| \\
 &= |\mathbf{c} \times \mathbf{b}| \\
 &= |\mathbf{c}| |\mathbf{b}| \sin \theta \\
 &= 6 |\mathbf{b}| \frac{1}{\sqrt{2}} \\
 \sqrt{2} \sqrt{2} &= 6 |\mathbf{b}| \\
 |\mathbf{b}| &= \frac{2}{6} = \frac{1}{3}
 \end{aligned}$$

- (c) The cosine of the angle between the planes $2x - 2y - z = 1$ and $x + 2y + 2z = 2$ is .

Answer: $-4/9$

Solutions:

$$\cos \theta = \left| \frac{\langle 2, -2, -1 \rangle \cdot \langle 1, 2, 2 \rangle}{\sqrt{2^2 + 2^2 + 1^2} \sqrt{1^2 + 2^2 + 2^2}} \right| = \left| \frac{-4}{9} \right| = \frac{4}{9}$$

- (d) A particle is moving along the path given by $x(t) = -t^2$, $y(t) = t^3 - 6t$. The largest interval over which the motion is up and to the right is $I = (a, b)$, where a is and b is .

Solutions: The curve is moving up when $y' > 0$ and to the right when $x' > 0$ and $y' > 0$.

$$x' = -2t \Rightarrow x' > 0 \text{ when } t < 0 \quad (1)$$

$$y' = 3t^2 - 6 \Rightarrow y' > 0 \text{ when } t^2 > 2 \quad (2)$$

Thus both derivatives are positive over $(-\infty, -\sqrt{2})$. We need the largest interval, so it would not be correct to use $b = -2$ even though the motion is up and to the right for $t < -2$.

- (e) The position of a moving object is given by the curve $\mathbf{r}(t) = \langle 1 - t, t^2, 3 \rangle$ for $t \in \mathbb{R}$. The speed of the object is given by . A unit vector that gives the direction of motion at time t is $\mathbf{T} = \langle f(t), g(t), h(t) \rangle$, where $f =$, $g =$, $h =$.

Solutions: The speed is the magnitude of the velocity.

$$\mathbf{r}'(t) = \mathbf{v} = \langle -1, 2t, 0 \rangle \quad (3)$$

$$|\mathbf{r}'(t)| = \sqrt{1 + (2t)^2} = \sqrt{1 + 4t^2} \quad (4)$$

We can write the speed as $\sqrt{1 + 4t^2}$. We do not need to express the answer using any units. The unit tangent vector $\mathbf{T}$ gives the direction of motion at any time t .

$$\mathbf{T} = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} = \left\langle \frac{-1}{\sqrt{1 + 4t^2}}, \frac{2t}{\sqrt{1 + 4t^2}}, 0 \right\rangle \quad (5)$$

Therefore,

$$f = -1/\sqrt{1 + 4t^2} \quad (6)$$

$$g = 2t/\sqrt{1 + 4t^2} \quad (7)$$

$$h = 0 \quad (8)$$

No further simplification is needed.

- (f) The velocity of a particle at time t is given by $\mathbf{v}(t) = \langle 4t, \sin(t), 2e^{2t} \rangle$ and the position of the object when $t = 0$ is $\mathbf{r} = \langle 1, 1, 1 \rangle$, then $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, where $f(t) =$, $g(t) =$, $h(t) =$.

Answer: $\langle 2t^2 + 1, -\cos(t) + 2, e^{2t} \rangle$

Solutions: $\mathbf{r} = \int \mathbf{v}(t) dt = \langle 2t^2 + c_1, -\cos(t) + c_2, e^{2t} + c_3 \rangle$. And $\mathbf{r}(0) = \langle c_1, c_2 - 1, 1 + c_3 \rangle = \langle 1, 1, 1 \rangle$. Therefore: $\mathbf{r} = \langle 2t^2 + 1, -\cos(t) + 2, e^{2t} \rangle$.

- (g) The unit tangent vector for a curve $\mathbf{r}(t)$ is $\mathbf{T}(t) = \langle 0, \cos t, \sin t \rangle$. The unit normal vector is $\mathbf{N} = \langle f(t), g(t), h(t) \rangle$, where $f(t) = \boxed{}$, $g(t) = \boxed{}$, $h(t) = \boxed{}$.

Answer: Solutions: The normal vector is

$$\mathbf{N} = \frac{d\mathbf{T}/dt}{|d\mathbf{T}/dt|} \quad (9)$$

And

$$\mathbf{T}'(t) = \langle 0, -\sin t, \cos t \rangle \quad (10)$$

$$|\mathbf{T}'(t)| = \sqrt{0 + (-\sin t)^2 + (\cos t)^2} = 1 \quad (11)$$

$$\mathbf{N} = \frac{d\mathbf{T}/dt}{|d\mathbf{T}/dt|} = \langle 0, -\sin(t), \cos(t) \rangle \quad (12)$$

3. (2 points) Indicate **true** if the statement is true, indicate **false** otherwise. You do not need to justify your reasoning.

true false

-
- a) $\vec{u} \times \vec{v} = \vec{v} \times \vec{u}$ for any vectors $\vec{u}, \vec{v} \in \mathbb{R}^3$. **Solution:** This is false. In fact, $\vec{u} \times \vec{v} = -\vec{v} \times \vec{u}$ making the cross product an anti-commutative operation. To see this note that while the parallelogram spanned by $\vec{u}$ and $\vec{v}$ doesn't change if their order is spanned which means that the length of the cross product won't change, the right hand rule does flip. Try sticking the pointer finger of your right hand straight out in any direction, rotating out your middle finger from this direction by any amount, and then popping up your thumb perpendicular to the previous two. Notice, that to flip the directions of your pointer and middle finger so that each goes to where the previous one was pointing (while keeping your pointer finger pointing straight out), you must flip your entire hand over and this flips the direction your thumb sticks out. Another perspective involves recalling the determinant notational trick for computing the cross product. Notice, that flipping the order of $\vec{u}$ and $\vec{v}$ flips two rows of the relevant matrix negating its determinant. ☐ ☐
- b) If $\mathbf{r}(t) = \langle t^2, 0 \rangle$ for $t \geq 0$, and $\mathbf{r}(t) = \langle 0, t^2 \rangle$ when $t < 0$, then $\mathbf{r}(t)$ is smooth. **Solution:** This is false. This function has a continuous derivative $\mathbf{r}'(t)$ which is $\langle 2t, 0 \rangle$ and $\langle 0, 2t \rangle$ when t is nonnegative/negative respectively. And note, these cases both converge to $\langle 0, 0 \rangle$ at $t = 0$ ensuring continuity. However, this means at $t = 0$ the derivative is the zero vector which is also a requirement of smoothness. Notice, this path traces down the positive y -axis before slowing to stop, turning 90 degrees, and then traces the positive x -axis going rightward at an increasing rate. We see that by allowing the parametrized point to slow to a stop it was able to trace a sharp corner without violating the continuity requirement. This is why this nonzero $\mathbf{r}'(t)$ condition is included in the definition of a smooth traced out curve. ☐ ☐
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4. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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5. (5 points) Suppose R is the point $R(0, 1, 2)$, L is the line $\mathbf{r}(t) = \langle t, t, t \rangle$. Please show your work.

- (a) Construct the equation of the plane that passes through R and contains the line L .

Solutions: Two points on the line are $O(0, 0, 0)$ and $Q(1, 1, 1)$. So two vectors parallel to the plane are $\mathbf{OR} = \langle 0, 1, 2 \rangle$ and $\mathbf{OQ} = \langle 1, 1, 1 \rangle$. A normal vector to the plane is

$$\mathbf{OR} \times \mathbf{OQ} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{vmatrix} = \langle -1, 2, -1 \rangle$$

The plane has equation

$$-1(x - 0) + (2)(y - 1) + (-1)(z - 2) = 0$$

Note: The above is sufficient but we could also rearrange this equation to obtain, for example, $-x + 2y - z = 0$. Note also that **it is not sufficient** to leave the answer as $-1(x - 0) + (2)(y - 1) + (-1)(z - 2)$, because this is not an equation. The question asked for an equation, not an expression. We need to have an equals sign somewhere in our answer.

- (b) Calculate the distance between the point R and the line L .

Solutions: A point on the line that corresponds to where $t = 0$ is $O(0, 0, 0)$. The line has direction vector $\mathbf{v} = \langle 1, 1, 1 \rangle$. The distance d can be obtained with the formula

$$d = \frac{|\mathbf{OR} \times \mathbf{v}|}{|\mathbf{v}|} \quad (13)$$

And

$$\mathbf{OR} = \langle 0, 1, 2 \rangle \quad (14)$$

$$\mathbf{OR} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{vmatrix} = \langle -1, 2, -1 \rangle \quad (15)$$

$$|\mathbf{OR} \times \mathbf{v}| = \sqrt{1 + 4 + 1} = \sqrt{6} \quad (16)$$

$$|\mathbf{v}| = \sqrt{1 + 1 + 1} = \sqrt{3} \quad (17)$$

$$d = \frac{|\mathbf{OR} \times \mathbf{v}|}{|\mathbf{v}|} = \sqrt{2} \quad (18)$$

6. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID: , the High School you attend: .

7. (5 points) An object is moving along a path given by $\mathbf{r}(t)$ for $t \geq 0$, where $\mathbf{r}'(t) = \mathbf{v} = \langle 4, t, 2\sqrt{2t} \rangle$. Please show your work for the parts below. You can use your results for each part using results from previous parts of this question.

- (a) Determine the speed, acceleration vector, and unit tangent vector for the curve.

$$s = |\mathbf{v}| = \sqrt{4^2 + t^2 + (2\sqrt{2t})^2} = \sqrt{t^2 + 8t + 16} = \sqrt{(t+4)^2} = t+4 \quad (19)$$

$$\frac{d}{dt} 2\sqrt{2t} = 2 \left(\frac{d}{dt} \sqrt{2t} \right) = 2 \cdot \frac{1}{2} (2t)^{-1/2} \frac{d}{dt} (2t) = \frac{2}{\sqrt{2t}} \quad (20)$$

$$\mathbf{a} = \mathbf{v}'(t) = \langle 0, 1, \frac{2}{\sqrt{2t}} \rangle = \mathbf{j} + \frac{2}{\sqrt{2t}} \mathbf{k} \quad (21)$$

$$\mathbf{T} = \frac{\mathbf{v}}{s} = \frac{1}{t+4} \langle 4, t, 2\sqrt{2t} \rangle = \frac{4}{t+4} \mathbf{i} + \frac{t}{t+4} \mathbf{j} + \frac{2\sqrt{2t}}{t+4} \mathbf{k} \quad (22)$$

- (b) Determine the length of the curve, L , from the point where $t = 4$ to $t = 8$. Simplify your answer as much as possible.

Apply the arc length formula.

$$L = \int_4^8 |\mathbf{v}| dt = \int_4^8 t+4 dt = \left(\frac{t^2}{2} + 4t \right) \Big|_4^8 = (32 + 32) - (8 + 16) = 64 - 24 = 40 \quad (23)$$

The question asked to simplify your work as much as possible, so for full credit you would not want to leave your answer as something like $64 - 24$.

- (c) Suppose the object starts at the origin, so that $\mathbf{r}(0) = \mathbf{0}$. Determine the value of t that corresponds to the point on the curve that is a distance of 10 units away from the origin along the curve. In other words, determine the value of t so that the arc length between the origin and $\mathbf{r}(t)$ is 10.

We can use the arc length parameter $s(t) = \int_{t_0}^t |\mathbf{v}| d\tau$, with $s = 10$, $t_0 = 0$, and solve for t .

$$s(t) = \int_{t_0}^t |\mathbf{v}| d\tau = 10 = \int_0^t \tau + 4 d\tau \quad (24)$$

$$10 = \left(\frac{\tau^2}{2} + 4\tau \right) \Big|_0^t \quad (25)$$

$$10 = \frac{t^2}{2} + 4t \quad (26)$$

$$0 = t^2 + 8t - 20 \quad (27)$$

$$0 = (t+10)(t-2) \quad (28)$$

We should ignore $t = -10$ because we were given that $t \geq 0$. So the value of t is $t = 2$.

Solutions

3.2 Year 1 MATH 2551 Exam 1 Sample B

Work done on scratch paper will not be graded. Please justify your reasoning on all questions unless indicated otherwise. Good luck on the exam!

A Few Helpful Formulas

$$d = \left| \mathbf{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|, \quad d = \frac{|\mathbf{PS} \times \mathbf{v}|}{|\mathbf{v}|}, \quad \mathbf{N} = \frac{d\mathbf{T}'(t)/dt}{|d\mathbf{T}/dt|}, \quad \kappa = \frac{|\mathbf{T}'(t)|}{|\mathbf{v}|} = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}, \quad \kappa = \frac{|f''(x)|}{[1 + (f'(x))^2]^{3/2}}$$

1. (0.5 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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2. (7 points) Fill in the blanks. You do not need to justify your reasoning.

- (a) The point on the sphere $(x - 2)^2 + (y - 4)^2 + (z - 7)^2 = 4$ nearest to the xy -plane is $P = (a, b, c)$, where $a = \boxed{}$, $b = \boxed{}$, $c = \boxed{}$. The radius of the sphere is $r = \boxed{}$.

Answer: $a = 2$, $b = 4$, $c = 5$, $r = 2$.

Solutions: The sphere has center $(2, 4, 7)$ and radius $r = 2$. Because the radius is 2 and the center has z -coordinate $z = 7$, the sphere lies above the xy -plane. The point on the bottom of the sphere closest to the xy -plane will be 2 units directly below the center. The coordinate is $(2, 4, 5)$, so $a = 2$, $b = 4$, $c = 5$.

- (b) The projection of $\mathbf{u} = \langle 2, 1, 2 \rangle$ onto the vector $\mathbf{v} = \langle 1, 0, 1 \rangle$ is $\text{proj}_{\mathbf{v}} \mathbf{u} = \langle a_1, a_2, a_3 \rangle$, where $a_1 = \boxed{}$, $a_2 = \boxed{}$, $a_3 = \boxed{}$. A vector in the direction of $\mathbf{u}$ with length 18 is $\mathbf{w} = \langle c_1, c_2, c_3 \rangle$ where $c_1 = \boxed{}$, $c_2 = \boxed{}$, $c_3 = \boxed{}$.

Solutions:

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v} = \frac{4}{2} \mathbf{v} = 2\mathbf{v}$$

So $a_1 = 2$, $a_2 = 0$, $a_3 = 2$. And a unit vector in the direction of $\mathbf{u}$ is

$$\hat{\mathbf{u}} = \frac{1}{3} \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \tag{29}$$

So a vector with length 18 in the direction of $\mathbf{u}$ is

$$18\hat{\mathbf{u}} = \frac{18}{3} \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} = 6 \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 12 \\ 6 \\ 12 \end{pmatrix} \tag{30}$$

So $c_1 = 12$, $c_2 = 6$, $c_3 = 12$.

- (c) The distance between the plane $x + 4y + 8z = 1$ and the point $S(4, 0, 0)$ is .

Answer: $1/3$

Solutions: A normal to the plane is $\mathbf{n} = \langle 1, 4, 8 \rangle$, and $|\mathbf{n}| = \sqrt{1^2 + 4^2 + 8^2} = \sqrt{81} = 9$. A point on the plane can be found by setting $y = z = 0$ and solving for x . Doing so gives us the point $P(1, 0, 0)$. Then $\mathbf{PS} = \langle 3, 0, 0 \rangle$. The distance is

$$d = \left| \mathbf{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right| = \frac{1}{9} \langle 3, 0, 0 \rangle \cdot \langle 1, 4, 8 \rangle = \frac{1}{3}. \quad (31)$$

- (d) Identify the surface $x - y^2 - 2y = 4z^2$ by type (cone, ellipsoid, etc.). .

Answer: elliptical paraboloid, or paraboloid

Solutions: In the plane $x = k$ for constant k , the curves are ellipses. But in the plane $y = 0$, we have a parabola $x = 4z^2$. So we can call this surface a paraboloid or elliptical paraboloid. Note the following.

- It is more accurate to refer to this shape as an elliptical paraboloid but we don't need to write paraboloid. It is sufficient to refer to this surface as a paraboloid.
- We don't need to write elliptical paraboloid because many people refer to $y = x^2$ and $y = 2x^2$ as parabolas. So, for us, it isn't necessary to indicate that the surface is an **elliptical** paraboloid.
- But be careful: **a hyperbolic paraboloid is not a paraboloid!** Do not refer to hyperbolic paraboloid as a paraboloid!

- (e) The tangent line to the curve $\mathbf{r}(t) = \langle 1 + t, t^2 \rangle$, $t \in \mathbb{R}$, passes through the point $P(1, -4)$ when $t = \text{$ and when $t = \text{$. Note that $\mathbf{r}$ does not pass through P .

Solutions: The tangent line that passes through the curve at $t = t_0$ is given by

$$\mathbf{L}(u) = u\mathbf{v}(t_0) + \mathbf{r}(t_0)$$

The line is parameterized by the variable u . Different values of u will give us different points on the tangent line at t_0 .

$$\mathbf{v}(t) = \mathbf{r}'(t) = \langle 1, 2t \rangle \quad (32)$$

$$\mathbf{r}(t_0) = \langle 1 + t_0, t_0^2 \rangle \quad (33)$$

$$\mathbf{L}(u) = u\mathbf{v} + \mathbf{r}(t_0) = \langle u + 1 + t_0, 2tu + t_0^2 \rangle \quad (34)$$

If the line passes through $P(1, -4)$, then there is a u and a t_0 so that

$$\langle u + 1 + t_0, 2tu + t_0^2 \rangle = \langle 1, -4 \rangle \quad (35)$$

In other words, we can solve

$$\begin{pmatrix} u + 1 + t_0 \\ 2tu + t_0^2 \end{pmatrix} = \begin{pmatrix} 1 \\ -4 \end{pmatrix} \quad (36)$$

Solving the first components gives us $u = -t_0$. The second component is

$$2tu + t_0^2 = -4 \quad (37)$$

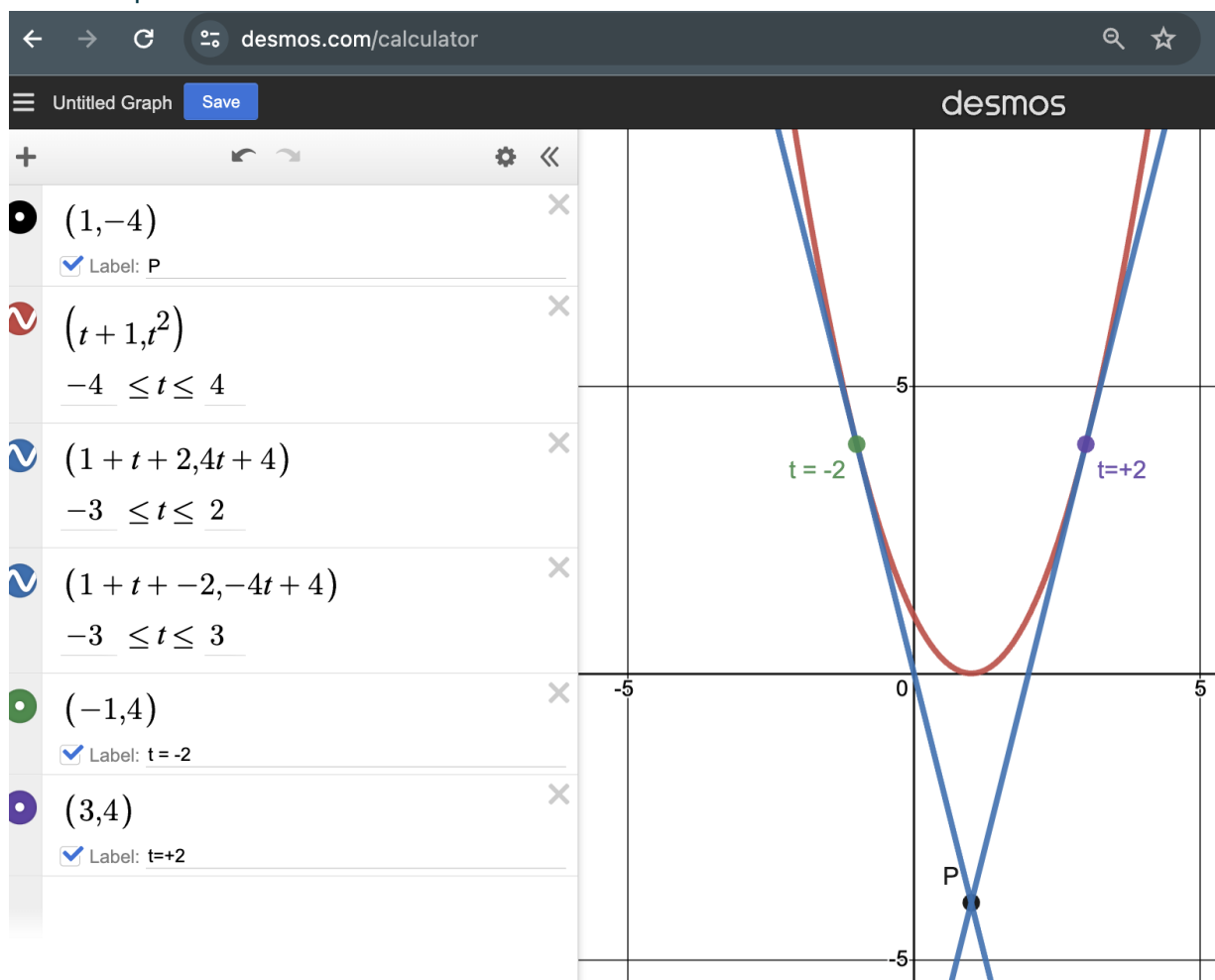
$$2t(-t_0) + t_0^2 = -4 \quad (38)$$

$$-2t_0^2 + t_0^2 = -4 \quad (39)$$

$$t_0^2 = 4 \quad (40)$$

$$t_0 = \pm 2 \quad (41)$$

The tangent lines that pass through P are tangent to the curve when $t = -2$ and $t = 2$. Sketching wasn't necessary for this problem but if we throw everything into Desmos this is what the problem looks like.



- (f) A projectile is launched from the ground with an initial speed of 120 m/sec, at an angle of elevation of θ , where $\sin \theta = \frac{1}{2}$. If we use $g = 10 \text{ m/s}^2$ as the acceleration due to gravity, the object will reach its maximum height when $t = \boxed{}$ seconds.

Solutions: Integrating $\mathbf{a}(t) = -g\mathbf{j}$ yields:

$$\mathbf{a}(t) = -g\mathbf{j} \Rightarrow \mathbf{v} = -gt\mathbf{j} + \mathbf{v}_0 \quad (42)$$

The initial velocity is

$$\mathbf{v}_0 = 120 \cos \theta \mathbf{i} + 120 \sin \theta \mathbf{j}$$

The object is at its maximum height when the $\mathbf{j}$ component of $\mathbf{v}$ is zero. Setting the $\mathbf{j}$ component to zero, with $\sin \theta = 1/2$ and $g = 10$, we find that

$$-gt + 120 \sin \theta = 0 \quad (43)$$

$$-10t + \frac{120}{2} = 0 \quad (44)$$

$$10t = 60 \quad (45)$$

$$t = 6 \quad (46)$$

So the object reaches its maximum height when $t = 6$ seconds.

- (g) The velocity vector of an object moving on the curve $\mathbf{r}(t)$ is $\mathbf{v}(t) = \langle 4 \cos(2t), 4 \sin(2t), 3 \rangle$ for $t > 0$. The speed of the object for $t > 0$ is $s(t) = |\mathbf{v}| = \boxed{}$. The unit tangent vector for $t > 0$ is $\mathbf{T} = \langle f(t), g(t), h(t) \rangle$, where $f(t) = \boxed{}$, $g(t) = \boxed{}$, $h = \boxed{}$. The curvature is $\kappa = \boxed{}$.

Answer: Solutions: The speed is the magnitude of the given velocity vector and is

$$s = |\mathbf{v}| = \sqrt{(4 \cos 2t)^2 + (4 \sin 2t)^2 + 3^2} = \sqrt{4^2(\cos^2 2t + \sin^2 2t) + 3^2} = \sqrt{16 + 9} = 5 \quad (47)$$

The unit tangent vector is

$$\mathbf{T} = \frac{\mathbf{v}}{|\mathbf{v}|} = \left\langle \frac{4}{5} \cos(2t), \frac{4}{5} \sin(2t), \frac{3}{5} \right\rangle \quad (48)$$

For curvature we also need

$$\frac{d\mathbf{T}}{dt} = \left\langle -\frac{8}{5} \sin 2t, \frac{8}{5} \cos 2t, 0 \right\rangle \quad (49)$$

$$\left| \frac{d\mathbf{T}}{dt} \right| = \sqrt{\left(-\frac{8}{5} \sin 2t \right)^2 + \left(\frac{8}{5} \cos 2t \right)^2} = \frac{8}{5} \quad (50)$$

The curvature is

$$\kappa = \frac{1}{|\mathbf{v}|} \left| \frac{d\mathbf{T}}{dt} \right| = \frac{1}{5} \cdot \frac{8}{5} = \frac{8}{25} \quad (51)$$

3. (2 points) Indicate **true** if the statement is true, indicate **false** otherwise. You do not need to justify your reasoning.

true false

-
- a) $(\vec{u} \times \vec{v}) \times \vec{w} = \vec{u} \times (\vec{v} \times \vec{w})$ for any vectors $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^3$. **Solution:** This is false. ☐ ☐
 The cross product is not an associative operation. For example, notice if $\vec{u} = \vec{v}$, then $\vec{u} \times \vec{v} = \vec{0}$ since they are parallel. This means $(\vec{u} \times \vec{v}) \times \vec{w} = \vec{0}$ while $\vec{u} \times (\vec{v} \times \vec{w})$ likely isn't nonzero. Try $\vec{u} = \vec{v} = \mathbf{i}$ and $\vec{w} = \mathbf{j}$ for a concrete example.
 With this knowledge, a nice exercise would be to consider what

$$(\vec{u} \times \vec{v}) \times \vec{w} - \vec{u} \times (\vec{v} \times \vec{w})$$

evaluates to in general. The formula

$$\vec{u} \times (\vec{v} \times \vec{w}) = (\vec{u} \cdot \vec{w})\vec{v} - (\vec{u} \cdot \vec{v})\vec{w}$$

will be very helpful with this task. Afterwards, look up “Jacobi Identity” and compare what you find with your results.

- b) Two lines in $\mathbb{R}^n$ that are parallel to a third line in $\mathbb{R}^3$ must be parallel. **Solution:** ☐ ☐
 This is true. Parallel lines have direction vectors in parametric vector form that are multiples of each other. The direction vectors of all three lines must be linearly dependent. This exercise was one of the problems from the Webwork homework.
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4. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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, the High School you attend: .
5. (5 points) Consider the point $S(4, 1, 1)$ and the line L whose parametric equations are $x = 2 + t$, $y = 1 - 3t$, $z = t$. Please show your work for the following.

- (a) Calculate the distance between S and L .

Solutions: A point on the line that corresponds to where $t = 0$ is $P(2, 1, 0)$. The line has direction vector $v = \langle 1, -3, 1 \rangle$. The distance d can be obtained with the formula

$$d = \frac{|\mathbf{PS} \times v|}{|v|} \quad (52)$$

And

$$\mathbf{PS} = \langle 4, 1, 1 \rangle - \langle 2, 1, 0 \rangle = \langle 2, 0, 1 \rangle \quad (53)$$

$$\mathbf{PS} \times v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 0 & 1 \\ 1 & -3 & 1 \end{vmatrix} = \langle 3, -1, -6 \rangle \quad (54)$$

$$|\mathbf{PS} \times v| = \sqrt{9 + 1 + 36} = \sqrt{46} \quad (55)$$

$$|v| = \sqrt{1 + 9 + 1} = \sqrt{11} \quad (56)$$

$$d = \frac{|\mathbf{PS} \times v|}{|v|} = \sqrt{\frac{46}{11}} \quad (57)$$

- (b) Determine the point P where L intersects the plane $x + y + 4z = 1$. Your answer should be expressed as a point in $\mathbb{R}^3$.

Solutions: The line will intersect the plane when it satisfies $x + y + 4z = 1$. Substituting the parametric equations for the line into the plane equation we obtain

$$(2 + t) + (1 - 3t) + 4(t) = 1 \quad (58)$$

$$3 + 2t = 1 \quad (59)$$

$$t = -1 \quad (60)$$

With $t = -1$, the parametric equations give us

$$x = 2 - 1 = 1 \quad (61)$$

$$y = 1 - 3(-1) = 4 \quad (62)$$

$$z = -1 \quad (63)$$

The point is $P(1, 4, -1)$.

6. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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7. (5 points)

An object is moving along a path given by $\mathbf{r}(t) = \langle t, \cos t, \sin t \rangle$ for $t \in \mathbb{R}$. Please show your work for the parts below. You can use your results for each part using results from previous parts of this question.

- (a) Determine the velocity, acceleration, and unit tangent vectors for the curve.

$$\mathbf{v} = \mathbf{r}'(t) = \mathbf{i} - \sin t \mathbf{j} + \cos t \mathbf{k} \quad (64)$$

$$|\mathbf{v}| = \sqrt{2} \quad (65)$$

$$\mathbf{T} = \frac{1}{\sqrt{2}} \langle 1, -\sin t, \cos t \rangle \quad (66)$$

$$\mathbf{a} = \mathbf{v}'(t) = \langle 0, -\cos t, -\sin t \rangle \quad (67)$$

- (b) Determine the length of the curve, L , from the point where $t = 4$ to $t = 12$.

Apply the arc length formula.

$$|\mathbf{v}| = \sqrt{2} \quad (68)$$

$$L = \int_4^{12} |\mathbf{v}| dt = \int_4^{12} \sqrt{2} dt = \sqrt{2} t \Big|_4^{12} = \sqrt{2}(12 - 4) = 8\sqrt{2} \quad (69)$$

For full credit do not leave your answer as $\sqrt{2} t \Big|_4^{12}$.

- (c) Determine the curvature of $\mathbf{r}(t)$ when $t = 0$.

There are a few different formulas for curvature. Let's use the formula

$$\kappa = \frac{1}{|\mathbf{v}|} \left| \frac{d\mathbf{T}}{dt} \right| \quad (70)$$

Then

$$|\mathbf{v}| = \sqrt{2} \quad (71)$$

$$\frac{d\mathbf{T}}{dt} = \frac{1}{\sqrt{2}} \langle 0, -\cos t, -\sin t \rangle \quad (72)$$

$$\left| \frac{d\mathbf{T}}{dt} \right| = \frac{1}{\sqrt{2}} \quad (73)$$

$$\kappa = \frac{1}{|\mathbf{v}|} \left| \frac{d\mathbf{T}}{dt} \right| = \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2}} = \frac{1}{2} \quad (74)$$

Note that there are formulas for curvature that are in sections of the textbook that we didn't cover. You aren't expected to know them for your exam, but if you like you can use them.

One such formula is $\kappa = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}$:

$$\mathbf{v} \times \mathbf{a} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -\sin t & \cos t \\ 0 & -\cos t & -\sin t \end{vmatrix} = \langle 1, \sin t, -\cos t \rangle \quad (75)$$

$$|\mathbf{v} \times \mathbf{a}| = \sqrt{2} \quad (76)$$

$$\kappa = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3} = \frac{\sqrt{2}}{(\sqrt{2})^3} = \frac{1}{2} \quad (77)$$

The curvature is $\frac{1}{2}$ for all t .

Solutions

3.3 Year 1 MATH 2551 Exam 1 Sample C

Work done on scratch paper will not be graded. Please justify your reasoning on all questions unless indicated otherwise. Good luck on the exam!

A Few Helpful Formulas

$$d = \left| \mathbf{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|, \quad d = \frac{|\mathbf{PS} \times \mathbf{v}|}{|\mathbf{v}|}, \quad \mathbf{N} = \frac{d\mathbf{T}'(t)/dt}{|d\mathbf{T}/dt|}, \quad \kappa = \frac{|\mathbf{T}'(t)|}{|\mathbf{v}|} = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}, \quad \kappa = \frac{|f''(x)|}{[1 + (f'(x))^2]^{3/2}}$$

1. (0.5 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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, the High School you attend: .
2. (7 points) Fill in the blanks. You do not need to justify your reasoning.

- (a) An equation of the plane that is perpendicular to the x -axis and passes through the point $P(3, 4, 5)$ is .

Answer: The plane has equation

$$\vec{n} \cdot (\vec{x} - \vec{x}_0) = 0 \tag{78}$$

Where $\vec{n}$ is a vector normal to the plane, $\vec{x}_0$ is any point in the plane, and $\vec{x}$ is the variable vector. We can use:

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \cdot \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} \right) = 0 \tag{79}$$

$$x - 3 = 0 \tag{80}$$

$$x = 3 \tag{81}$$

- (b) Suppose $\mathbf{u} \times \mathbf{v} = \langle 2, 3, 4 \rangle$ and $(2\mathbf{u} + 3\mathbf{v}) \times \mathbf{v} = \mathbf{w}$. Then $\mathbf{w} = \langle c_1, c_2, c_3 \rangle$ where $c_1 = \input{width=1cm,height=1cm}, c_2 = \input{width=1cm,height=1cm}, c_3 = \input{width=1cm,height=1cm}$.

Solutions:

$$\begin{aligned} \mathbf{w} &= (2\mathbf{u} + 3\mathbf{v}) \times \mathbf{v} \\ &= 2\mathbf{u} \times \mathbf{v} + 3\mathbf{v} \times \mathbf{v} \\ &= 2\mathbf{u} \times \mathbf{v} \\ &= 2\langle 2, 3, 4 \rangle \\ &= \langle 4, 6, 8 \rangle \end{aligned}$$

So $c_1 = 4$, $c_2 = 6$, $c_3 = 8$.

- (c) The plane that passes through $P(5, 0, 2)$ and contains the line $x = 1 + t$, $y = t$, $z = 2 + t$ is .

Solutions: Plane is parallel to direction vector of given line, $\mathbf{v} = \langle 1, 1, 1 \rangle$. Line contains $Q(1, 0, 2)$, so plane also parallel to the vector $\mathbf{PQ} = \langle 1, 0, 2 \rangle - \langle 5, 0, 2 \rangle = \langle -4, 0, 0 \rangle$. It would also be ok to use any scalar multiple of this.

Unit normal to plane is

$$\mathbf{n} = \mathbf{v} \times \mathbf{PQ} = \begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ -4 & 0 & 0 \end{vmatrix} = \langle 0, -4, 4 \rangle$$

So using point P and $\mathbf{n}$, the plane has equation

$$(0)(x - 5) + (-4)(y - 0) + (4)(z - 2) = 0 \quad (82)$$

which could be simplified to

$$y - z = 2$$

It isn't necessary to simplify the equation further. But we could also express the answer as

$$y - z + 2 = 0 \quad (83)$$

- (d) Identify the surface $x^2 - y^2 + z = 0$ by type (cone, ellipsoid, etc.). .

Answer: saddle, or hyperbolic paraboloid. It would not be correct to refer to this shape as a paraboloid. A paraboloid is a very different surface. But we could refer to this surface as a Pringle chip because it looks like one. Try entering this into Desmos 3D: $x^2 - y^2 + z = 0 \{x^2 + y^2 \leq 1\}$.

- (e) The equation of the tangent line to the curve $\mathbf{r}(t) = \langle 1 - t, t^2, 3t \rangle$ at the point where $t = 1$ has equation $\mathbf{L} = \langle f(t), g(t), h(t) \rangle$, where $f(t) = \text{}$, $g(t) = \text{}$, $h(t) = \text{}$.

Solutions: The tangent line is parallel to $\mathbf{r}'(t)$.

$$\mathbf{r}'(t) = \langle -1, 2t, 3 \rangle \quad (84)$$

$$\mathbf{r}'(1) = \langle -1, 2, 3 \rangle \quad (85)$$

And $\mathbf{r}(1) = \langle 0, 1, 3 \rangle$. Therefore, $\mathbf{L}(t) = \langle 0 - t, 1 + 2t, 3 + 3t \rangle$.

- (f) If the velocity of an object at time t is $\mathbf{v}(t) = \langle 6t^2, \cos(t), 4e^{2t} \rangle$ and its position at $t = 0$ is $\mathbf{r} = \langle 1, 2, 0 \rangle$, then $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, where $f(t) = \text{}$, $g(t) = \text{}$, $h(t) = \text{}$.

Solutions: Integrating $\mathbf{r}(t)$:

$$\mathbf{r} = \int \mathbf{v}(t) dt = \langle 2t^3 + c_1, \sin(t) + c_2, 2e^{2t} + c_3 \rangle \quad (86)$$

And $\mathbf{r}(0) = \langle c_1, c_2, 2 + c_3 \rangle = \langle 1, 2, 1 \rangle$. Therefore, $\mathbf{r} = \langle 2t^3 + 1, 2 + \sin(t), 2e^{2t} - 2 \rangle$.

- (g) A plane curve $\mathbf{r}(t)$ has curvature $\kappa = \frac{1}{8}$ at $\mathbf{r}(t_0) = \langle 15, 2 \rangle$, and the unit normal vector at $t = t_0$ is $\mathbf{N} = \mathbf{j} = \langle 0, 1 \rangle$. The equation of the osculating circle at $t = t_0$ is .

Answer: Solutions: Recall that The circle of curvature, or osculating circle, at a point P on a plane curve where $\kappa \neq 0$ is the circle in the plane of the curve that

- i. is tangent to the curve at P (has the same tangent line the curve has)
- ii. has the same curvature the curve has at P
- iii. lies toward the concave or inner side of the curve

The **radius of curvature** of the curve at P is the radius of the circle of curvature, which is $\frac{1}{\kappa}$. So to obtain the radius, we calculate κ and take the reciprocal. The **center** of the osculating circle of the curve at P lies on the inner side of the curve, and the unit normal points in the direction of the inner side of the planar curve.

So for this problem we have a circle with radius $1/\kappa = 8$, and whose centre is 8 units away from the point $(15, 2)$ in the direction of $\mathbf{N}$. The circle has equation

$$(x - 15)^2 + (y - 10)^2 = 8^2$$

3. (2 points) Indicate **true** if the statement is true, indicate **false** otherwise. You do not need to justify your reasoning.

true false

-
- a) If $\vec{u} \neq \vec{0}$ and $\vec{v} \neq \vec{0}$ are nonparallel vectors in $\mathbb{R}^3$, then $(c_1\vec{u} + c_2\vec{v}) \cdot (\vec{u} \times \vec{v}) = 0$ only if $c_1 = c_2 = 0$. **Solution:** This is false. This expression will be zero regardless of the values of c_1 and c_2 . Recall that $c_1\vec{u} + c_2\vec{v}$ will always lie in the plane spanned by $\vec{u}$ and $\vec{v}$, and recall $\vec{u} \times \vec{v}$ is always perpendicular to this plane. So, these vectors are always orthogonal and their dot product is zero. ☐ true ☐ false
- b) If $\mathbf{r}(t) = \langle t^2, 2t^2 \rangle$ for $t \in \mathbb{R}$, then $\mathbf{r}(t)$ is smooth. **Solution:** This is false. This function has a continuous derivative $\mathbf{r}'(t)$ which is $\langle 2t, 4t \rangle$ when t is nonnegative/negative respectively. And note that the derivative converges to $\langle 0, 0 \rangle$ at $t = 0$ ensuring continuity. However, this means at $t = 0$ the derivative is the zero vector which is also a requirement of smoothness. Note that the curve moves towards the origin along a straight line as t increases from negative infinity to the origin. Then the curve **doubles back** across its path as t increases towards positive infinity from the origin. Smoothness requires that the curve does not double back on itself. ☐ true ☐ false
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4. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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, the High School you attend: .

5. (5 points) A particle is located at the point $P(1, 2, 1)$ at time $t = 0$. It travels in a straight line to the point $Q(5, 6, 8)$, has speed 18 at $t = 0$, and acceleration $\mathbf{a} = 4\mathbf{i} + 4\mathbf{j} + 7\mathbf{k}$. Determine the velocity of the particle, $\mathbf{v}(t)$. Please show your work.

Answer: $\langle 4t + 8, 4t + 8, 7t + 14 \rangle$

Solutions: The velocity $\mathbf{v}$ is given by the integral of the acceleration, so

$$\mathbf{v} = \int \mathbf{a}(t) dt = \langle 4t, 4t, 7t \rangle + \mathbf{c}_1 \quad (87)$$

The speed at $t = 0$ is 18 and the particle is moving in the direction of $\langle 4, 4, 7 \rangle$, so

$$\mathbf{v}(0) = \frac{18}{\sqrt{16 + 16 + 49}} \langle 4, 4, 7 \rangle = \frac{18}{\sqrt{81}} \langle 4, 4, 7 \rangle = \mathbf{c}_1 \quad (88)$$

$$\langle 8, 8, 14 \rangle = \mathbf{c}_1 \quad (89)$$

Thus

$$\mathbf{v} = \langle 4t, 4t, 7t \rangle + \langle 8, 8, 14 \rangle = \langle 4t + 8, 4t + 8, 7t + 14 \rangle \quad (90)$$

6. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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7. (5 points)

An object is moving in the xy -plane along a path given by $\mathbf{r}(t) = \langle t, t^2 + 1 \rangle$ for $t \in \mathbb{R}$. Please show your work for the parts below. You can use your results for each part using results from previous parts of this question.

- (a) Determine the velocity, acceleration vector, and unit tangent vector for the curve.

$$\mathbf{v} = \mathbf{r}'(t) = \mathbf{i} + 2t\mathbf{j} \quad (91)$$

$$\mathbf{a} = \mathbf{v}'(t) = 2\mathbf{j} \quad (92)$$

$$\mathbf{T} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{1}{\sqrt{4t^2 + 1}} (\mathbf{i} + 2t\mathbf{j}) \quad (93)$$

- (b) Determine the curvature of $\mathbf{r}(t)$ when $t = 0$.

A planar curve of the form $y = f(x)$ can be parameterized using $\mathbf{r}(x) = x\mathbf{i} + f(x)\mathbf{j}$, and in such a case it can be shown that we can compute the curvature using

$$\kappa = \frac{|f''(x)|}{[1 + (f'(x))^2]^{3/2}}$$

For this problem, we can parameterize the curve by setting $x = t$ and $f = f(y) = f(x(t)) = t^2 + 1$, and use this curvature formula as follows.

$$\kappa(t) = \frac{|f''(t)|}{(1 + (f'(t))^2)^{3/2}} = \frac{2}{(1 + (2t)^2)^{3/2}} \quad (94)$$

$$\kappa(0) = \frac{2}{(1 + 0)^{3/2}} = \frac{2}{1^{3/2}} = 2 \quad (95)$$

Note that it is also possible to calculate curvature in other ways, as there are other formulas for curvature.

- (c) Given that the principal normal vector when $t = 0$ is $\mathbf{N} = \mathbf{j}$, determine the equation of the osculating circle when $t = 0$.

The circle has radius $1/\kappa = \frac{1}{2}$ and touches the curve at the point where $t = 0$. This corresponds to the point $(0, 1)$, because $\mathbf{r}(t) = \langle t, t^2 + 1 \rangle$, and

$$\mathbf{r}(0) = \langle 0, 1 \rangle \quad (96)$$

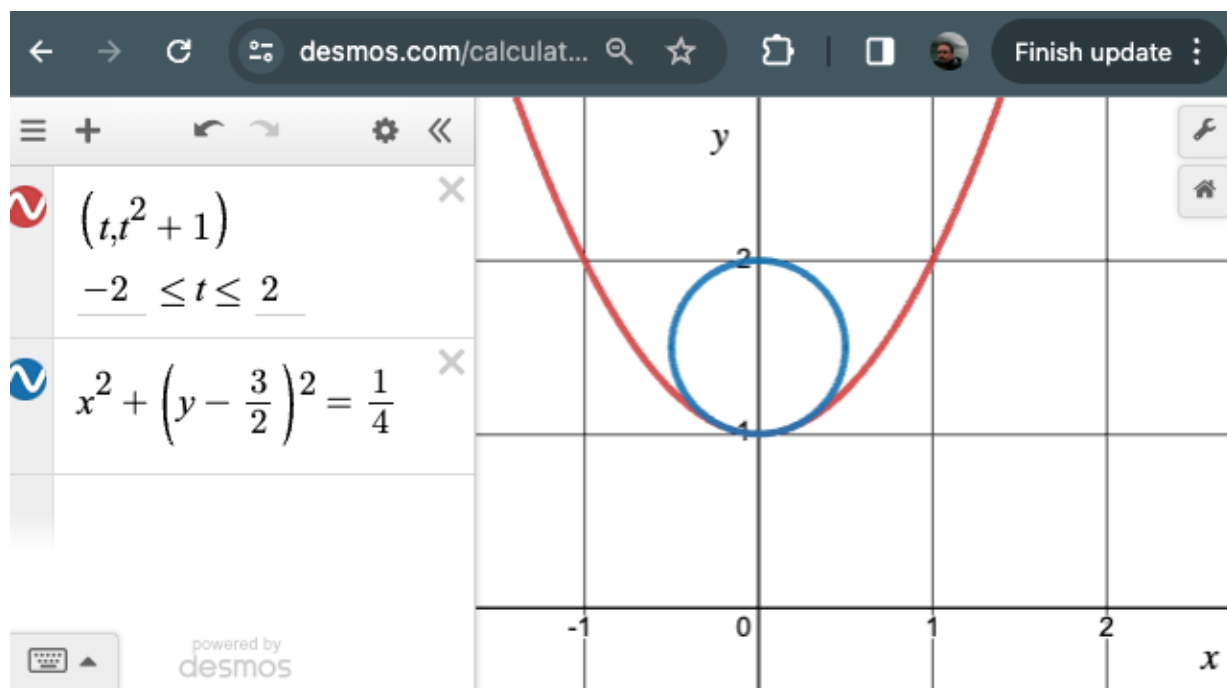
The normal vector points in the direction of $\mathbf{j}$, so the circle has center that is $\frac{1}{2}$ units above the point $(0, 1)$, which is the point $(0, 1 + \frac{1}{2})$. The circle has the equation

$$(x - 0)^2 + \left(y - \frac{3}{2}\right)^2 = \left(\frac{1}{2}\right)^2$$

We could also write this as

$$x^2 + \left(y - \frac{3}{2}\right)^2 = \frac{1}{4}$$

Sketching wasn't necessary but if we put the curve and the circle into Desmos, the picture below is what we get. Note that the circle touches the curve at the point $(0, 1)$, that it lies above the curve, and has radius $1/2$.



Solutions

3.4 Year 1 MATH 2551 Exam 1 Sample D

Work done on scratch paper will not be graded. Please justify your reasoning on all questions unless indicated otherwise. Good luck on the exam!

A Few Helpful Formulas

$$d = \left| \mathbf{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|, \quad d = \frac{|\mathbf{PS} \times \mathbf{v}|}{|\mathbf{v}|}, \quad \mathbf{N} = \frac{d\mathbf{T}'(t)/dt}{|d\mathbf{T}/dt|}, \quad \kappa = \frac{|\mathbf{T}'(t)|}{|\mathbf{v}|} = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}, \quad \kappa = \frac{|f''(x)|}{[1 + (f'(x))^2]^{3/2}}$$

1. (0.5 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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2. (7 points) Fill in the blanks. You do not need to justify your reasoning.

- (a) The distance between the plane $x + 2y + 2z = 2$ and the point $S(8, 2, 1)$ is . The distance between S and the yz -plane is .

Solutions: A normal to the plane is $\mathbf{n} = \langle 1, 2, 2 \rangle$, and $|\mathbf{n}| = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{9} = 3$. A point on the plane can be found by setting $y = z = 0$ and solving for x . Doing so gives us the point $P(2, 0, 0)$. Then $\mathbf{PS} = \langle 6, 2, 1 \rangle$.

$$d = \left| \mathbf{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right| = \frac{1}{3} \langle 6, 2, 1 \rangle \cdot \langle 1, 2, 2 \rangle = \frac{12}{3} = 4 \quad (97)$$

The point S is 8 units away from the yz -plane because the point has coordinates $(8, 2, 1)$. So the distance between S and the yz -plane is 8.

- (b) The parallelogram determined by vectors $\mathbf{a}$ and $\mathbf{b}$ has area 6. If $\mathbf{a} = \mathbf{b} + \mathbf{c}$, and $\mathbf{c}$ makes an angle of $\frac{\pi}{2}$ with $\mathbf{b}$, and $|\mathbf{c}| = 2$, then $|\mathbf{b}| = \text{}$.

Solutions:

$$\begin{aligned} 6 &= |\mathbf{a} \times \mathbf{b}| \\ &= |(\mathbf{b} + \mathbf{c}) \times \mathbf{b}| \\ &= |\mathbf{b} \times \mathbf{b} + \mathbf{c} \times \mathbf{b}| \\ &= |\mathbf{c} \times \mathbf{b}| \\ &= |\mathbf{c}| |\mathbf{b}| \sin \theta \\ &= 2 |\mathbf{b}| 1 \\ 6 &= 2 |\mathbf{b}| \\ |\mathbf{b}| &= 3 \end{aligned}$$

- (c) The plane that passes through $P(5, 0, 2)$ and contains the line $x = 1 + 4t$, $y = t$, $z = 2 + 3t$ is .

Solutions: Plane is parallel to direction vector of given line, $\mathbf{v} = \langle 4, 1, 3 \rangle$. Line contains $Q(1, 0, 2)$, so plane also parallel to the vector $\mathbf{PQ} = \langle 1, 0, 2 \rangle - \langle 5, 0, 2 \rangle = \langle -4, 0, 0 \rangle$. It would also be ok to use any scalar multiple of this.

Unit normal to plane is

$$\mathbf{n} = \mathbf{v} \times \mathbf{PQ} = \begin{vmatrix} i & j & k \\ 4 & 1 & 3 \\ -4 & 0 & 0 \end{vmatrix} = \langle 0, -12, 4 \rangle$$

So using point P and $\mathbf{n}$, the plane has equation

$$(0)(x - 5) + (-12)(y - 0) + (4)(z - 2) = 0 \quad (98)$$

which could be simplified to

$$3y - z = -2$$

It isn't necessary to simplify the equation.

- (d) Identify the surface $x - y^2 + z^2 = 0$ by type (cone, ellipsoid, etc.). .

Answer: saddle, or hyperbolic paraboloid. It would not be correct to refer to this shape as a paraboloid. A paraboloid is a very different surface.

- (e) The position of a moving object is given by the curve $\mathbf{r}(t) = \langle 1 - t^2, t^3 \rangle$ for $t \in \mathbb{R}$. The value of t that corresponds to the time when the direction of motion is parallel to $\mathbf{w} = \langle -2, 6 \rangle$, is $t =$.

Solutions: The direction of motion is parallel to the velocity.

$$\mathbf{r}'(t) = \mathbf{v} = \langle -2t, 3t^2 \rangle \quad (99)$$

Vectors are parallel when they are non-zero multiples of each other. So the velocity will be parallel to the given $\mathbf{w}$ vector when they are non-zero multiples of each other. If we let $\lambda \neq 0$ be the multiple, then we can solve

$$\mathbf{v} = \lambda \mathbf{w} \quad (100)$$

$$\begin{pmatrix} -2t \\ 3t^2 \end{pmatrix} = \lambda \begin{pmatrix} -2 \\ 6 \end{pmatrix} \quad (101)$$

The first components give us $t = \lambda$. The second components give us

$$3t^2 = 6\lambda \quad (102)$$

But $\lambda = t$, so we have

$$3t^2 = 6t \quad (103)$$

$$t(t - 2) = 0 \quad (104)$$

So the values of t that satisfy $\mathbf{v} = \lambda \mathbf{w}$ are $t = 0$ and $t = 2$. But $\mathbf{r}(0)$ is not parallel to $\mathbf{w}$. So the only value of t where the two vectors are parallel is $t = 2$.

- (f) Consider the initial value problem (IVP): $\mathbf{r}' = 32t\mathbf{i} + 6t^2\mathbf{j} + 4\mathbf{k}$, $\mathbf{r}(0) = 3\mathbf{i} + 5\mathbf{j} + 8\mathbf{k}$. The solution to the IVP is $\mathbf{r} = \langle f(t), g(t), h(t) \rangle$, where $f(t) = \boxed{}$, $g(t) = \boxed{}$, $h(t) = \boxed{}$.

Solutions:

$$f(t) = \int 32t \, dt = 16t^2 + c_1 \quad (105)$$

$$g(t) = \int 6t^2 \, dt = 2t^3 + c_2 \quad (106)$$

$$h(t) = \int 4 \, dt = 4t + c_3 \quad (107)$$

Then using $\mathbf{r}(0)$, we find $c_1 = 3$, $c_2 = 5$, $c_3 = 8$. So

$$f(t) = 16t^2 + 3 \quad (108)$$

$$g(t) = 2t^3 + 5 \quad (109)$$

$$h(t) = 4t + 8 \quad (110)$$

- (g) The velocity vector for an object moving on the curve $\mathbf{r}(t)$ is $\mathbf{v}(t) = \langle t \cos t, t \sin t \rangle$. The speed of the object for $t > 0$ is $s(t) = \boxed{}$. The unit tangent vector for $t > 0$ is $\mathbf{T} = \langle f(t), g(t) \rangle$, where $f(t) = \boxed{}$, $g(t) = \boxed{}$.

Answer: Solutions: The speed is the magnitude of the velocity vector and is

$$s = |\mathbf{v}| = \sqrt{(t \cos t)^2 + (t \sin t)^2} = \sqrt{t^2(\cos^2 t + \sin^2 t)} = |t| \quad (111)$$

But we are told that $t > 0$ so we can use $|t| = t$. The unit tangent vector is

$$\mathbf{T} = \frac{\mathbf{v}}{|\mathbf{v}|} = \langle \cos t, \sin t \rangle \quad (112)$$

3. (2 points) Indicate **true** if the statement is true, indicate **false** otherwise. You do not need to justify your reasoning.

true false

-
- a) $|(\vec{u} \times \vec{v}) \cdot \vec{w}|$, $|(\vec{v} \times \vec{w}) \cdot \vec{u}|$, and $|(\vec{w} \times \vec{u}) \cdot \vec{v}|$ can all evaluate to distinct values for some triple of vectors in $\mathbb{R}^3$. **Solution:** This is false. Recall the absolute value of the triple vector product $|(\vec{u} \times \vec{v}) \cdot \vec{w}|$ measures the volume of the parallelepiped spanned by the vectors $\vec{u}$, $\vec{v}$, and $\vec{w}$. Reordering these vectors doesn't change the shape spanned so all of these expressions must evaluate to the same value. Using the determinant based definition of the cross product try proving that all three expression evaluate to the same value even if we drop the absolute values. ☐ true ☐ false
- b) Two lines in $\mathbb{R}^3$ that are orthogonal to a third line in $\mathbb{R}^3$ are parallel to each other. **Solution:** False. The x -axis and the y -axis are orthogonal to the z -axis. But the x -axis and the y -axis are not parallel. This exercise was one of the problems from the Webwork homework. ☐ true ☐ false
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4. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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5. (5 points) Consider the two planes P_1 and P_2 .

$$P_1 : 2x + 3y + 6z = 2$$

$$P_2 : 4x + 6y + 12z = 6$$

Please show your work for the following.

- (a) Calculate the distance between planes P_1 and P_2 .

Solutions: The planes are parallel because their normal vectors are parallel. A point on the first plane is $P(1, 0, 0)$, a point on the other plane is $S(0, 1, 0)$, then $\mathbf{PS} = \langle -1, 1, 0 \rangle$, and a normal vector to both planes is

$$\mathbf{n} = \langle 2, 3, 6 \rangle$$

The distance between that point and the other plane is the length of the projection of vector $\mathbf{PS}$ onto $\mathbf{n}$.

$$d = |\text{proj}_{\mathbf{n}} \mathbf{PS}| = \left| \frac{\mathbf{PS} \cdot \mathbf{n}}{\mathbf{n} \cdot \mathbf{n}} \mathbf{n} \right| \quad (113)$$

$$= \left| \frac{\langle -1, 1, 0 \rangle \cdot \langle 2, 3, 6 \rangle}{2^2 + 3^2 + 6^2} \mathbf{n} \right| \quad (114)$$

$$= \left| \frac{-2 + 3}{4 + 9 + 36} \mathbf{n} \right| \quad (115)$$

$$= \left| \frac{1}{49} \mathbf{n} \right| \quad (116)$$

$$= \frac{1}{49} |\mathbf{n}| \quad (117)$$

$$= \frac{1}{49} \sqrt{2^2 + 3^2 + 6^2} \quad (118)$$

$$= \frac{7}{49} \quad (119)$$

$$= 1/7 \quad (120)$$

The distance between the two planes is $1/7$.

Additional Solution Notes

Note that there are other formulas for the distance between a point and a plane that can be used. One such formula, which is in the textbook, is

$$d = \frac{|\mathbf{PS} \cdot \mathbf{n}|}{|\mathbf{n}|} \quad (121)$$

Yet another formula for the distance between parallel planes $ax + by + cz + d_1 = 0$ and $ax + by + cz + d_2 = 0$ is

$$d = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}} \quad (122)$$

Also note that a good way to check your work is to consider other points for P and Q and see if you get the same result for the distance.

- (b) Determine the parametric equations for the line of intersection between plane P_1 and the plane P_3 , which is given by $2x + 4y + 8z = 4$.

Solutions: the set of points that are common to P_1 and P_3 satisfy the linear system whose augmented matrix is

$$\begin{pmatrix} 2 & 3 & 6 & 2 \\ 2 & 4 & 8 & 4 \end{pmatrix} \sim \begin{pmatrix} 2 & 3 & 6 & 2 \\ 0 & 1 & 2 & 2 \end{pmatrix} \sim \begin{pmatrix} 2 & 0 & 0 & -4 \\ 0 & 1 & 2 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 2 & 2 \end{pmatrix} \quad (123)$$

So if the variable vector $\mathbf{x} = \langle x, y, z \rangle$ represents the line of intersection, then:

$$\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2 \\ 2 - 2z \\ z \end{pmatrix} \quad (124)$$

Letting $z = t$, we have the parametric equations

$$x = -2 \quad (125)$$

$$y = 2 - 2t \quad (126)$$

$$z = t \quad (127)$$

6. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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7. (5 points)

An object is moving in the xy -plane along a path given by $\mathbf{r}(t) = \langle t, ((t-4)^2+1) \rangle = \langle t, t^2-8t+17 \rangle$ for $t \geq 0$. Please show your work for the parts below. You can use your results for each part using results from previous parts of this question.

- (a) Determine the velocity and acceleration vectors for the curve for any $t \geq 0$.

$$\mathbf{v} = \mathbf{r}'(t) = \mathbf{i} + (2t - 8)\mathbf{j} \quad (128)$$

$$\mathbf{a} = \mathbf{v}'(t) = 2\mathbf{j} \quad (129)$$

- (b) Determine the curvature of $\mathbf{r}(t)$ when $t = 4$.

Setting $x = t$ and $f(x) = x^2 - 8x + 17$ we may calculate the curvature using the formula

$$\kappa = \frac{|f''(x)|}{(1 + (f'(x))^2)^{3/2}} = \frac{2}{(1 + (2x - 8)^2)^{3/2}} \quad (130)$$

$$\kappa(4) = \frac{2}{(1 + (2(4) - 8)^2)^{3/2}} = \frac{2}{1^{3/2}} = 2 \quad (131)$$

The above formula was introduced in one of the homework problems for Section 13.4. Note that it is also possible to calculate curvature in other ways, as there are several other formulas for curvature.

- (c) Determine the unit tangent vector $\mathbf{T}$ when $t = 4$.

Using the formulas for these vectors we obtain

$$|\mathbf{v}| = \sqrt{1 + (2t - 8)^2} \quad (132)$$

$$\mathbf{T} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{\langle 1, 2t - 8 \rangle}{\sqrt{1 + (2t - 8)^2}} \quad (133)$$

$$\mathbf{T}(4) = \langle 1, 0 \rangle \quad (134)$$

- (d) Given that the principal normal vector when $t = 4$ is $\mathbf{N} = \mathbf{j}$, determine the equation of the osculating circle when $t = 4$.

The circle has radius $1/\kappa = \frac{1}{2}$ and touches the curve at the point

$$\mathbf{r}(4) = \langle 4, 1 \rangle \quad (135)$$

The normal vector points in the direction of $\mathbf{j}$, so the circle has center $(4, 1 + \frac{1}{2})$. The circle has the equation $(x - 4)^2 + (y - 3/2)^2 = 1/4$.

Solutions

3.5 Year 1 MATH 2551 Exam 1 Sample E

Work done on scratch paper will not be graded. Please justify your reasoning on all questions unless indicated otherwise. Good luck on the exam!

A Few Helpful Formulas

$$d = \left| \mathbf{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|, \quad d = \frac{|\mathbf{PS} \times \mathbf{v}|}{|\mathbf{v}|}, \quad \mathbf{N} = \frac{d\mathbf{T}'(t)/dt}{|d\mathbf{T}/dt|}, \quad \kappa = \frac{|\mathbf{T}'(t)|}{|\mathbf{v}|} = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}, \quad \kappa = \frac{|f''(x)|}{[1 + (f'(x))^2]^{3/2}}$$

1. (0.5 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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2. (7 points) Fill in the blanks. You do not need to justify your reasoning.

- (a) The cosine of the angle between the vectors $\langle 4, 0, 3 \rangle$ and $\langle 2, 1, 2 \rangle$ is .

$$\text{Solutions: } \cos \theta = \frac{\langle 4, 0, 3 \rangle \cdot \langle 2, 1, 2 \rangle}{\sqrt{4^2 + 3^2} \sqrt{2^2 + 2^2 + 1}} = \frac{8 + 0 + 6}{\sqrt{25} \cdot \sqrt{9}} = \frac{14}{15}.$$

- (b) The parallelogram determined by vectors $\mathbf{a}$ and $\mathbf{b}$ has area 24. If $\mathbf{a} = \mathbf{b} + \mathbf{c}$, and $\mathbf{c}$ makes an angle of $\frac{\pi}{2}$ with $\mathbf{b}$, and $|\mathbf{c}| = 2$, then $|\mathbf{b}| =$.

Solutions:

$$\begin{aligned} 24 &= |\mathbf{a} \times \mathbf{b}| \\ &= |(\mathbf{b} + \mathbf{c}) \times \mathbf{b}| \\ &= |\mathbf{b} \times \mathbf{b} + \mathbf{c} \times \mathbf{b}| \\ &= |\mathbf{c} \times \mathbf{b}| \\ &= |\mathbf{c}| |\mathbf{b}| \sin \theta \\ &= 2 |\mathbf{b}| \cdot 1 \\ 24 &= 2 |\mathbf{b}| \\ |\mathbf{b}| &= 12 \end{aligned}$$

- (c) The plane that passes through $P(5, 0, 2)$ and contains the line $x = 1 + 4t$, $y = t$, $z = 2 + 2t$ is .

Solutions: Plane is parallel to direction vector of given line, $\mathbf{v} = \langle 4, 1, 2 \rangle$. Line contains $Q(1, 0, 2)$, so plane also parallel to the vector $\mathbf{PQ} = \langle 1, 0, 2 \rangle - \langle 5, 0, 2 \rangle = \langle -4, 0, 0 \rangle$. It would also be ok to use any scalar multiple of this.

Unit normal to plane is

$$\mathbf{n} = \mathbf{v} \times \mathbf{PQ} = \begin{vmatrix} i & j & k \\ 4 & 1 & 2 \\ -4 & 0 & 0 \end{vmatrix} = \langle 0, -8, 4 \rangle$$

So using point P and $\mathbf{n}$, the plane has equation

$$(0)(x - 5) + (-8)(y - 0) + (4)(z - 2) = 0 \quad (136)$$

which could be simplified to

$$2y - z = -2$$

It isn't necessary to simplify the equation.

- (d) Identify the surface $y + z^2 = 1 - x^2$ by type (cone, ellipsoid, etc.).

Answer: paraboloid. The vertex of the paraboloid is at the point $(0, 1, 0)$. It is ok to call this an elliptical paraboloid, because a paraboloid is a type of elliptical paraboloid. On the other hand, **it is not correct to call this a hyperbolic paraboloid**, because that is a very different surface!

- (e) The equation of the tangent line to the curve $\mathbf{r}(t) = \langle t - 1, 3t, t^2 \rangle$ at the point where $t = 2$ has equation $\mathbf{L} = \langle f(t), g(t), h(t) \rangle$, where $f(t) = \text{$, $g(t) = \text{$, $h(t) = \text{$.

Solutions: The tangent line is parallel to $\mathbf{r}'(t)$.

$$\mathbf{r}'(t) = \langle 1, 3, 2t \rangle \quad (137)$$

$$\mathbf{r}'(2) = \langle 1, 3, 4 \rangle \quad (138)$$

And $\mathbf{r}(2) = \langle 1, 6, 4 \rangle$. Therefore, $\mathbf{L}(t) = \langle 1 + t, 6 + 3t, 4 + 4t \rangle$.

- (f) If the velocity of an object at time t is $\mathbf{v}(t) = \langle 6t^2, \cos(t), 4e^{2t} \rangle$ and its position at $t = 0$ is $\mathbf{r} = \langle 1, 2, 1 \rangle$, then $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, where $f(t) = \text{$, $g(t) = \text{$, $h(t) = \text{$.

Solutions: Integrating $\mathbf{r}(t)$:

$$\mathbf{r} = \int \mathbf{v}(t) dt = \langle 2t^3 + c_1, \sin(t) + c_2, 2e^{2t} + c_3 \rangle \quad (139)$$

And $\mathbf{r}(0) = \langle c_1, c_2, 2 + c_3 \rangle = \langle 1, 2, 1 \rangle$. Therefore, $\mathbf{r} = \langle 2t^3 + 1, 2 + \sin(t), 2e^{2t} - 1 \rangle$.

- (g) A plane curve $\mathbf{r}(t)$ has curvature $\kappa = \frac{1}{4}$ at the point $\mathbf{r}(t_0) = 3\mathbf{i} + 5\mathbf{j}$, and the unit normal vector at t_0 is $\mathbf{N} = \mathbf{i} = \langle 1, 0 \rangle$. The equation of the osculating circle at $t = t_0$ is .

Answer: Solutions: Recall that The circle of curvature, or osculating circle, at a point P on a plane curve where $\kappa \neq 0$ is the circle in the plane of the curve that

- is tangent to the curve at P (has the same tangent line the curve has)
- has the same curvature the curve has at P
- lies toward the concave or inner side of the curve

The **radius of curvature** of the curve at P is the radius of the circle of curvature, which is $\frac{1}{\kappa}$. So to obtain the radius, we calculate κ and take the reciprocal. The **center** of the osculating circle of the curve at P lies on the inner side of the curve, and the unit normal points in the direction of the inner side of the planar curve.

So for this problem we have a circle with radius $1/\kappa = 4$, and whose centre is 4 units away from the point $(3, 5)$ in the direction of $\mathbf{N}$. The circle has equation

$$(x - 7)^2 + (y - 5)^2 = 4^2$$

3. (2 points) Indicate **true** if the statement is true, indicate **false** otherwise. You do not need to justify your reasoning.

true false

-
- | | | |
|--|-----------------------|-----------------------|
| a) It is possible to have more than n nonzero vectors in $\mathbb{R}^n$ which are all mutually orthogonal to each other. Solution: This is false. Recall from linear algebra that any mutually orthogonal set of vectors must be linearly independent and linearly independent sets can contain as many vectors as dimensions of the space. | ☐ | ☐ |
| b) Two lines in $\mathbb{R}^3$ are either parallel or they intersect each other. Solution: False. The lines may be skew. This exercise was one of the problems from the Webwork homework. | ☐ | ☐ |
-

4. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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5. (5 points) A particle is located at the point $P(1, 2, 5)$ at time $t = 0$. It travels in a straight line to the point $Q(2, 4, 7)$, has speed 12 at $t = 0$, and acceleration $\mathbf{a} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$. Determine the velocity of the particle, $\mathbf{v}(t)$. Please show your work.

Answer: $\langle t + 4, 2t + 8, 2t + 8 \rangle$

Solutions: The velocity $\mathbf{v}$ is given by the integral of the acceleration, so

$$\mathbf{v} = \int \mathbf{a}(t) dt = \langle t, 2t, 2t \rangle + \mathbf{c}_1 \quad (140)$$

The speed at $t = 0$ is 12 and the particle is moving in the direction of $\langle 1, 2, 2 \rangle$, so

$$\mathbf{v}(0) = \frac{12}{\sqrt{1+4+4}} \langle 1, 2, 2 \rangle = \mathbf{c}_1 \quad (141)$$

$$\langle 4, 8, 8 \rangle = \mathbf{c}_1 \quad (142)$$

Thus

$$\mathbf{v} = \langle t, 2t, 2t \rangle + \langle 4, 8, 8 \rangle = \langle t + 4, 2t + 8, 2t + 8 \rangle \quad (143)$$

6. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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7. (5 points)

An object is moving along a path given by $\mathbf{r}(t) = \langle t, \cos t, \sin t \rangle$ for $t \in \mathbb{R}$. Please show your work for the parts below. You can use your results for each part using results from previous parts of this question.

- (a) Determine the velocity, acceleration, and unit tangent vectors for the curve.

$$\mathbf{v} = \mathbf{r}'(t) = \mathbf{i} - \sin t \mathbf{j} + \cos t \mathbf{k} \quad (144)$$

$$|\mathbf{v}| = \sqrt{2} \quad (145)$$

$$\mathbf{T} = \frac{1}{\sqrt{2}} \langle 1, -\sin t, \cos t \rangle \quad (146)$$

$$\mathbf{a} = \mathbf{v}'(t) = \langle 0, -\cos t, -\sin t \rangle \quad (147)$$

- (b) Determine the length of the curve, L , from the point where $t = 1$ to $t = 4$.

Apply the arc length formula.

$$|\mathbf{v}| = \sqrt{2} \quad (148)$$

$$L = \int_1^4 |\mathbf{v}| dt = \int_1^4 \sqrt{2} dt = \sqrt{2} t \Big|_1^4 = \sqrt{2}(4 - 1) = 3\sqrt{2} \quad (149)$$

For full credit do not leave your answer as $\sqrt{2}t \Big|_1^4$.

- (c) Determine the curvature of $\mathbf{r}(t)$ when $t = 0$.

There are a few different formulas for curvature. Let's use the formula

$$\kappa = \frac{1}{|\mathbf{v}|} \left| \frac{d\mathbf{T}}{dt} \right| \quad (150)$$

Then

$$|\mathbf{v}| = \sqrt{2} \quad (151)$$

$$\frac{d\mathbf{T}}{dt} = \frac{1}{\sqrt{2}} \langle 0, -\cos t, -\sin t \rangle \quad (152)$$

$$\left| \frac{d\mathbf{T}}{dt} \right| = \frac{1}{\sqrt{2}} \quad (153)$$

$$\kappa = \frac{1}{|\mathbf{v}|} \left| \frac{d\mathbf{T}}{dt} \right| = \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2}} = \frac{1}{2} \quad (154)$$

Note that there are formulas for curvature that are in sections of the textbook that we didn't cover. You aren't expected to know them for your exam, but if you like you can use them.

One such formula is $\kappa = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}$:

$$\mathbf{v} \times \mathbf{a} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -\sin t & \cos t \\ 0 & -\cos t & -\sin t \end{vmatrix} = \langle 1, \sin t, -\cos t \rangle \quad (155)$$

$$|\mathbf{v} \times \mathbf{a}| = \sqrt{1^2 + \sin^2 t + \cos^2 t} = \sqrt{2} \quad (156)$$

$$\kappa = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3} = \frac{\sqrt{2}}{(\sqrt{2})^3} = \frac{1}{2} \quad (157)$$

Either way, the curvature is $\frac{1}{2}$ for all t .

Solutions

3.6 Year 1 MATH 2551 Exam 1 Sample F

Work done on scratch paper will not be graded. Please justify your reasoning on all questions unless indicated otherwise. Good luck on the exam!

A Few Helpful Formulas

$$d = \left| \mathbf{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|, \quad d = \frac{|\mathbf{PS} \times \mathbf{v}|}{|\mathbf{v}|}, \quad \mathbf{N} = \frac{d\mathbf{T}'(t)/dt}{|d\mathbf{T}/dt|}, \quad \kappa = \frac{|\mathbf{T}'(t)|}{|\mathbf{v}|} = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}, \quad \kappa = \frac{|f''(x)|}{[1 + (f'(x))^2]^{3/2}}$$

1. (0.5 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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, the High School you attend: .
2. (7 points) Fill in the blanks. You do not need to justify your reasoning.

- (a) The projection of the point $P(2, 3, 4)$ onto the yz -plane is the point $Q(c_1, c_2, c_3)$ where $c_1 = \boxed{}$, $c_2 = \boxed{}$, $c_3 = \boxed{}$.

Solutions: the closest point on the yz -plane to the point P will have the same y and z coordinates as P , and will have x -coordinate of zero. The point is $Q(0, 3, 4)$, so $c_1 = 0$, $c_2 = 3$, $c_3 = 4$. No need for projection formulas.

- (b) The cosine of the angle between the vectors $\mathbf{a} = \langle 2, 1, 2 \rangle$ and $\mathbf{b} = \langle 1, 0, 2 \rangle$ is $\cos \theta = \boxed{}$.

Solutions:

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} = \frac{2 + 4}{\sqrt{2^2 + 1^2 + 2^2} \sqrt{1^2 + 2^2}} = \frac{6}{\sqrt{9} \sqrt{5}} = \frac{6}{3\sqrt{5}} = \frac{2}{\sqrt{5}}$$

- (c) The distance between the plane $x + 4y + 8z = 1$ and the point $S(28, 0, 0)$ is .

Answer: 3

Solutions: A normal to the plane is $\mathbf{n} = \langle 1, 4, 8 \rangle$, and $|\mathbf{n}| = \sqrt{1^2 + 4^2 + 8^2} = \sqrt{81} = 9$. A point on the plane can be found by setting $y = z = 0$ and solving for x . Doing so gives us the point $P(1, 0, 0)$. Then $\mathbf{PS} = \langle 27, 0, 0 \rangle$. The distance is

$$d = \left| \mathbf{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right| = \frac{1}{9} \langle 27, 0, 0 \rangle \cdot \langle 1, 4, 8 \rangle = 3. \quad (158)$$

- (d) Identify the surface $-x^2 + y^2 + 4z^2 = 0$ by type (cone, paraboloid, etc.). .

Answer: elliptical cone

Solutions: The curve can be expressed as $x^2 = y^2 + 4z^2$. In the plane $x = k$ for constant k , the curve is an ellipse with equation $y^2 + 4z^2 = k$. The surface intersects the plane $z = 0$

along two straight lines $x = \pm y$, and likewise the plane $y = 0$ along the lines $z = \pm 4x$. Note that it isn't necessary to indicate that the surface is an **elliptical** cone. It is sufficient to refer to this surface as a cone. There is no such thing as a hyperbolic cone, at least in this course.

- (e) The tangent line to the curve $\mathbf{r}(t) = \langle 4t - 1, t^3, t^2 \rangle$ at the point where $t = 2$ is $\mathbf{L} = \langle f(t), g(t), h(t) \rangle$, where $f(t) = \boxed{}$, $g(t) = \boxed{}$, $h(t) = \boxed{}$.

Solutions: The tangent line is parallel to $\mathbf{r}'(t)$.

$$\mathbf{r}'(t) = \langle 4, 3t^2, 2t \rangle \quad (159)$$

$$\mathbf{r}'(2) = \langle 4, 12, 4 \rangle \quad (160)$$

And $\mathbf{r}(2) = \langle 7, 8, 4 \rangle$. Therefore, $\mathbf{L}(t) = \langle 7 + 4t, 8 + 12t, 4 + 4t \rangle$.

- (f) A projectile is launched from the ground with an initial speed of 500 m/sec at an angle of elevation of θ , where $\sin \theta = \frac{1}{2}$. If we use $g = 10 \text{ m/s}^2$ as the acceleration due to gravity, the object will return to the ground when $t = \boxed{}$ seconds.

Solutions: Integrating $\mathbf{a}(t) = -g\mathbf{j}$ yields:

$$\mathbf{a}(t) = -g\mathbf{j} \Rightarrow \mathbf{v} = -gt\mathbf{j} + \mathbf{v}_0 \quad (161)$$

$$\mathbf{r} = -\frac{1}{2}gt^2\mathbf{j} + \mathbf{v}_0t + \mathbf{r}_0 \quad (162)$$

But the initial position is $\mathbf{r}_0 = \mathbf{0}$, and the initial velocity is

$$\mathbf{v}_0 = 500 \cos \theta \mathbf{i} + 500 \sin \theta \mathbf{j}$$

The object is on the ground when the $\mathbf{j}$ component of $\mathbf{r}$ is zero. Setting the $\mathbf{j}$ component to zero and using $\sin \theta = 1/2$ and $g = 10$ gives us

$$-\frac{1}{2}gt^2 + 500 \sin \theta t = 0 \quad (163)$$

$$\left(-\frac{gt}{2} + 500 \sin \theta\right)t = 0 \quad (164)$$

$$\left(-\frac{10t}{2} + \frac{500}{2}\right)t = 0 \quad (165)$$

$$(-5t + 250)t = 0 \quad (166)$$

$$(-t + 50)t = 0 \quad (167)$$

So the object reaches the ground when $t = 50$ seconds.

- (g) The velocity vector for an object moving on the curve $\mathbf{r}(t)$ is $\mathbf{v}(t) = \langle 4t \cos t, 4t \sin t \rangle$ for $t > 0$. The speed of the object for $t > 0$ is $s(t) = |\mathbf{v}| = \boxed{}$. The unit tangent vector for $t > 0$ is $\mathbf{T} = \langle f(t), g(t) \rangle$, where $f(t) = \boxed{}$, $g(t) = \boxed{}$. The curvature is $\kappa = \boxed{}$.

Answer: Solutions: The speed is the magnitude of the given velocity vector and is

$$s = |\mathbf{v}| = \sqrt{(4t \cos t)^2 + (4t \sin t)^2} = \sqrt{4^2 t^2 (\cos^2 t + \sin^2 t)} = 4|t| \quad (168)$$

But we are told that $t > 0$ so we can assume $|t| = t$. The unit tangent vector is

$$\mathbf{T} = \frac{\mathbf{v}}{|\mathbf{v}|} = \langle \cos t, \sin t \rangle \quad (169)$$

We need

$$\frac{d\mathbf{T}}{dt} = \langle -\sin t, \cos t \rangle \quad (170)$$

$$\left| \frac{d\mathbf{T}}{dt} \right| = \sqrt{(-\sin t)^2 + \cos^2 t} = 1 \quad (171)$$

The curvature is

$$\kappa = \frac{1}{|\mathbf{v}|} \left| \frac{d\mathbf{T}}{dt} \right| = \frac{1}{4t} \quad (172)$$

3. (2 points) Indicate **true** if the statement is true, indicate **false** otherwise. You do not need to justify your reasoning.

true false

- a) The surface defined by the equation $y = x^2$ in $\mathbb{R}^3$ is an example of a cylinder. ☐ ☐

Solution: This is true. Recall that a cylinder is defined as any surface which can be generated by moving a line along a given planar curve. The curve is called a **generating curve** for the cylinder. In this example, the parabola $y = x^2$ exists in the xy -plane of $\mathbb{R}^3$. If we take the line along the z -axis and translate it to intersect this parabolic curve in the xy -plane at some point, then every other point on this line will also satisfy the equation $y = x^2$. Notice, the only difference between points of this line are their z -coordinates which aren't relevant to this equation! Conversely, every point in $\mathbb{R}^3$ which satisfies this equation is on such a line as we simply project this point down onto the xy -plane to see where to make the translation.

- b) The trace obtained from the intersection of the plane $z = 1$ and the surface $(x/2)^2 + (y/3)^2 + (z/4)^2 = 4$ is a circle. ☐ ☐
Solution: False. Substitution yields $(x/2)^2 + (y/3)^2 + (1/4)^2 = 4$, which is the equation of an ellipse. This exercise was one of the problems from the Webwork homework.

4. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID: , the High School you attend: .
5. (5 points) Consider the point $S(4, 1, 1)$ and the L_1 whose parametric equations are $x = 2 + t$, $y = 1 - t$, $z = t$. Please show your work for the following.

- (a) Calculate the distance between the point and line L_1 . *Solutions:* A point on the line that corresponds to where $t = 0$ is $P(2, 1, 0)$. The line has direction vector $v = \langle 1, -1, 1 \rangle$. The distance d can be obtained with the formula

$$d = \frac{|\mathbf{PS} \times v|}{|v|} \quad (173)$$

And

$$\mathbf{PS} = \langle 4, 1, 1 \rangle - \langle 2, 1, 0 \rangle = \langle 2, 0, 1 \rangle \quad (174)$$

$$\mathbf{PS} \times v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 0 & 1 \\ 1 & -1 & 1 \end{vmatrix} = \langle 1, -1, -2 \rangle \quad (175)$$

$$|\mathbf{PS} \times v| = \sqrt{1 + 1 + 4} = \sqrt{6} \quad (176)$$

$$|v| = \sqrt{1 + 1 + 1} = \sqrt{3} \quad (177)$$

$$d = \frac{|\mathbf{PS} \times v|}{|v|} = \sqrt{\frac{6}{3}} = \sqrt{2} \quad (178)$$

- (b) Construct the component equation of the plane that contains L_1 and the line L_2 whose parametric equations are $x = 2 + 2t$, $y = 1 + 4t$, $z = 3t$.

Solutions: Direction vectors for the lines are:

$$L_1: \quad \mathbf{v} = \langle 1, -1, 1 \rangle \quad (179)$$

$$L_2: \quad \mathbf{u} = \langle 2, 4, 3 \rangle \quad (180)$$

The plane contains the direction vectors of both lines, and a vector perpendicular to them is

$$\mathbf{n} = \mathbf{v} \times \mathbf{u} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 1 \\ 2 & 4 & 3 \end{vmatrix} = \langle -7, -1, 6 \rangle \quad (181)$$

A point in both lines is $P(2, 1, 0)$. The plane has component equation

$$-7(x - 2) - (y - 1) + 6z = 0 \quad (182)$$

It is not necessary to simplify further.

6. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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7. (5 points)

An object is moving along a path given by $\mathbf{r}(t)$ for $t \geq 0$, where $\mathbf{r}'(t) = \mathbf{v} = \langle t, 3, \sqrt{6t} \rangle$. Please show your work for the parts below. You can use your results for each part using results from previous parts of this question.

- (a) Determine the speed, acceleration vector, and unit tangent vector for the curve.

$$s = |\mathbf{v}| = \sqrt{t^2 + 3^2 + (\sqrt{6t})^2} = \sqrt{t^2 + 6t + 9} = \sqrt{(t+3)^2} = t+3 \quad (183)$$

$$\mathbf{a} = \mathbf{v}'(t) = \langle 1, 0, 3t^{-1/2} \rangle = \mathbf{i} + \frac{3}{\sqrt{6t}}\mathbf{k} \quad (184)$$

$$\mathbf{T} = \frac{\mathbf{v}}{s} = \frac{1}{t+3} \langle t, 3, \sqrt{6t} \rangle = \frac{t}{t+3}\mathbf{i} + \frac{3}{t+3}\mathbf{j} + \frac{\sqrt{6t}}{t+3}\mathbf{k} \quad (185)$$

- (b) Determine the length of the curve, L , from the point where $t = 2$ to $t = 4$.

Apply the arc length formula.

$$L = \int_2^4 |\mathbf{v}| dt = \int_2^4 t+3 dt = \left(\frac{t^2}{2} + 3t \right) \Big|_2^4 = (8+12) - (2+6) = 12 \quad (186)$$

- (c) Suppose the object starts at the origin, so that $\mathbf{r}(0) = \mathbf{0}$. Determine the value of t that corresponds to the point on the curve that is a distance of 8 units away from the origin along the curve. In other words, determine the value of t so that the arc length between the origin and $\mathbf{r}(t)$ is 8.

We can use the arc length parameter.

$$s(t) = \int_{t_0}^t |\mathbf{v}| d\tau \quad (187)$$

We set $s = 8$, $t_0 = 0$, and solve for t .

$$s(t) = \int_{t_0}^t |\mathbf{v}| d\tau = 8 = \int_0^t \tau + 3 d\tau \quad (188)$$

$$8 = \left(\frac{\tau^2}{2} + 3\tau \right) \Big|_0^t \quad (189)$$

$$8 = \frac{t^2}{2} + 3t \quad (190)$$

$$0 = t^2 + 6t - 16 \quad (191)$$

$$0 = (t+8)(t-2) \quad (192)$$

We should ignore $t = -8$ because we were given that $t \geq 0$. So the value of t is $t = 2$.

Solutions

3.7 Year 1 MATH 2551 Exam 1 Sample G

Work done on scratch paper will not be graded. Please justify your reasoning on all questions unless indicated otherwise. Good luck on the exam!

A Few Helpful Formulas

$$d = \left| \mathbf{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|, \quad d = \frac{|\mathbf{PS} \times \mathbf{v}|}{|\mathbf{v}|}, \quad \mathbf{N} = \frac{d\mathbf{T}'(t)/dt}{|d\mathbf{T}/dt|}, \quad \kappa = \frac{|\mathbf{T}'(t)|}{|\mathbf{v}|} = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}, \quad \kappa = \frac{|f''(x)|}{[1 + (f'(x))^2]^{3/2}}$$

- (0.5 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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, the High School you attend: .
- (7 points) Fill in the blanks. You do not need to justify your reasoning.

- The point on the sphere $(x - 2)^2 + (y - 4)^2 + (z - 6)^2 = 4$ nearest to the yz -plane is $P = (a, b, c)$, where $a = \text{$, $b = \text{$, $c = \text{$.

Answer: $a = 0, b = 4, c = 6$.

Solutions: The sphere has center $(2, 4, 6)$ and radius 2. Because the radius is 2 and the center has x -coordinate $x = 2$, the sphere lies to the right of the yz -plane. The point on the sphere closest to the xy -plane will be 2 units directly from the center. The coordinate is $(0, 4, 6)$, so $a = 0, b = 4, c = 6$.

- Given the points $P(2, 0, 1), Q(3, 1, 1), R(2, 0, 2)$, the area of triangle PQR is .

Answer: $\sqrt{2}/2$.

Solutions:

$$\text{area} = \frac{1}{2} |PQ \times PR| = \frac{1}{2} \left| \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \right| = \frac{\sqrt{1^2 + (-1)^2 + 0}}{2} = \sqrt{2}/2$$

The area is $\sqrt{2}/2$.

- The distance between the plane $x + 2y + 2z = 2$ and the point $S(8, 2, 1)$ is . The distance between S and the yz -plane is .

Solutions: A normal to the plane is $\mathbf{n} = \langle 1, 2, 2 \rangle$, and $|\mathbf{n}| = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{9} = 3$. A point on the plane can be found by setting $y = z = 0$ and solving for x . Doing so gives us the point $P(2, 0, 0)$. Then $\mathbf{PS} = \langle 6, 2, 1 \rangle$.

$$d = \left| \mathbf{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right| = \frac{1}{3} \langle 6, 2, 1 \rangle \cdot \langle 1, 2, 2 \rangle = \frac{12}{3} = 4 \quad (193)$$

The point S is 8 units away from the yz -plane because the point has coordinates $(8, 2, 1)$. So the distance between S and the yz -plane is 8.

- (d) Identify the surface $x^2 + 2z^2 = 16 + 4y - y^2$ by type (cone, ellipsoid, etc.).

Answer: ellipsoid

Solutions: Rearrange and complete the square.

$$\begin{aligned}x^2 + 2z^2 &= 16 + 4y - y^2 \\16 &= y^2 - 4y + x^2 + 2z^2 \\16 &= y^2 - 4y + 4 - 4 + x^2 + 2z^2 \\16 &= (y - 2)^2 - 4 + x^2 + 2z^2 \\20 &= (y - 2)^2 + x^2 + 2z^2\end{aligned}$$

Not a sphere because coefficient in front of squared terms not all the same. Surface is an ellipsoid. We can refer to spheres as ellipsoids (because a sphere is a special case of an ellipsoid), but we can't refer to an ellipsoid as a sphere unless the ellipsoid is a sphere.

- (e) The position of a moving object is given by the curve $\mathbf{r}(t) = \langle 2 + t, 4, t^2 \rangle$ for $t \in \mathbb{R}$. The speed of the object at $t = 2$ is . A unit vector that gives the direction of motion at time $t = 2$ is $\mathbf{T} = \langle f(t), g(t), h(t) \rangle$, where $f = \langle \text{input} \rangle$, $g = \langle \text{input} \rangle$, $h = \langle \text{input} \rangle$.

Solutions: The speed is the magnitude of the velocity.

$$\mathbf{r}'(t) = \mathbf{v} = \langle 1, 0, 2t \rangle \quad (194)$$

$$|\mathbf{r}'(t)| = \sqrt{1 + 4t^2} \quad (195)$$

$$|\mathbf{r}'(2)| = \sqrt{1 + 16} = \sqrt{17} \quad (196)$$

We can write the speed as $\sqrt{17}$. We do not need to express the answer using any units. The unit tangent vector $\mathbf{T}$ gives the direction of motion at any time t .

$$\mathbf{T} = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} = \left\langle \frac{1}{\sqrt{1 + 4t^2}}, 0, \frac{2t}{\sqrt{1 + 4t^2}} \right\rangle \quad (197)$$

Therefore,

$$f(2) = \frac{1}{\sqrt{1 + 4(2)^2}} = \frac{1}{\sqrt{17}} \quad (198)$$

$$g(2) = 0 \quad (199)$$

$$h(2) = \frac{2(2)}{\sqrt{1 + 4(2)^2}} = \frac{4}{\sqrt{17}} \quad (200)$$

No further simplification needed. But it is also ok to write the answer as

$$f(2) = \frac{\sqrt{17}}{17} \quad (201)$$

$$g(2) = 0 \quad (202)$$

$$h(2) = \frac{4\sqrt{17}}{17} \quad (203)$$

- (f) A projectile is launched from the ground with an initial speed of 80 m/sec, at an angle of elevation of θ , where $\sin \theta = \frac{1}{4}$. If we use $g = 10 \text{ m/s}^2$ as the acceleration due to gravity, the object will reach its maximum height when $t = \boxed{}$ seconds.

Solutions: Integrating $\mathbf{a}(t) = -g\mathbf{j}$ yields:

$$\mathbf{a}(t) = -g\mathbf{j} \Rightarrow \mathbf{v} = -gt\mathbf{j} + \mathbf{v}_0 \quad (204)$$

The initial velocity is

$$\mathbf{v}_0 = 80 \cos \theta \mathbf{i} + 80 \sin \theta \mathbf{j}$$

The object is at its maximum height when the $\mathbf{j}$ component of $\mathbf{v}$ is zero. Setting the $\mathbf{j}$ component to zero, with $\sin \theta = 1/4$ and $g = 10$, we find that

$$-gt + 80 \sin \theta = 0 \quad (205)$$

$$-10t + \frac{80}{4} = 0 \quad (206)$$

$$10t = 20 \quad (207)$$

$$t = 2 \quad (208)$$

So the object reaches its maximum height when $t = 2$ seconds.

- (g) The velocity vector of an object moving on the curve $\mathbf{r}(t)$ is $\mathbf{v}(t) = \langle 3 \cos(2t), 3 \sin(2t) \rangle$ for $t > 0$. The speed of the object for $t > 0$ is $s(t) = |\mathbf{v}| = \boxed{}$. The unit tangent vector for $t > 0$ is $\mathbf{T} = \langle f(t), g(t) \rangle$, where $f(t) = \boxed{}$, and $g(t) = \boxed{}$. The curvature for $t > 0$ is $\boxed{}$.

Answer: Solutions: The speed is the magnitude of the given velocity vector and is

$$s = |\mathbf{v}| = \sqrt{(3 \cos t)^2 + (3 \sin t)^2} = \sqrt{3^2(\cos^2 t + \sin^2 t) + 3^2} = \sqrt{9} = 3 \quad (209)$$

The unit tangent vector is

$$\mathbf{T} = \frac{\mathbf{v}}{|\mathbf{v}|} = \left\langle \frac{3}{3} \cos(2t), \frac{3}{3} \sin(2t) \right\rangle = \langle \cos(2t), \sin(2t) \rangle \quad (210)$$

For curvature we also need

$$\frac{d\mathbf{T}}{dt} = \langle -2 \sin 2t, 2 \cos 2t \rangle \quad (211)$$

$$\left| \frac{d\mathbf{T}}{dt} \right| = \sqrt{(-2 \sin 2t)^2 + (2 \cos 2t)^2} = 2 \quad (212)$$

The curvature is

$$\kappa = \frac{1}{|\mathbf{v}|} \left| \frac{d\mathbf{T}}{dt} \right| = \frac{1}{3} \cdot 2 = \frac{2}{3} \quad (213)$$

3. (2 points) Indicate **true** if the statement is true, indicate **false** otherwise. You do not need to justify your reasoning.

true false

-
- a) The surface defined by the equation $z^2 = x^2 + y^2$ is an example of a cylinder. ☐ ☐

Solution: This is false. Notice, this is the equation of a double-sided cone. Indeed if we square root both sides, then we obtain $z = \pm\sqrt{x^2 + y^2}$ which by Pythagoras Theorem means we have the set of points which are the same distance above or below the xy -plane as they are from the z -axis. This is not a cylinder, because though it is full of lines, no union of purely translated offset lines yields the cone.

To convincingly argue this, notice that every straight line of the cone must go through the vertex at the origin. Otherwise, it would either intersect the xy plane at a point other than the origin clearly violating $z^2 = x^2 + y^2$ or its a horizontal line which can't fit within the circular horizontal cross sections of a cone. With one point fixed, we see that no two unique lines in our cone go in the same direction and translation alone will not sweep out the cone. Hence, this surface is not a cylinder.

- b) The surface $5x^2 + 8y^2 - z = 0$ is an elliptical paraboloid whose axis of symmetry is the z -axis. ☐ ☐
-
- Solution:** True.

4. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID: , the High School you attend: .
5. (5 points) Consider the point $S(4, 3, 1)$ and the L_1 whose parametric equations are $x = 2 + t$, $y = 1 - 2t$, $z = 2t$. Please show your work for the following.

- (a) Calculate the distance between the point and line L_1 . *Solutions:* A point on the line that corresponds to where $t = 0$ is $P(2, 1, 0)$. The line has direction vector $\mathbf{v} = \langle 1, -2, 2 \rangle$. The distance d can be obtained with the formula

$$d = \frac{|\mathbf{PS} \times \mathbf{v}|}{|\mathbf{v}|} \quad (214)$$

And

$$\mathbf{PS} = \langle 4, 3, 1 \rangle - \langle 2, 1, 0 \rangle = \langle 2, 2, 1 \rangle \quad (215)$$

$$\mathbf{PS} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 2 & 1 \\ 1 & -2 & 2 \end{vmatrix} = \langle 6, -3, -6 \rangle \quad (216)$$

$$|\mathbf{PS} \times \mathbf{v}| = \sqrt{36 + 36 + 9} = \sqrt{81} = 9 \quad (217)$$

$$|\mathbf{v}| = \sqrt{1 + 4 + 4} = \sqrt{9} = 3 \quad (218)$$

$$d = \frac{|\mathbf{PS} \times \mathbf{v}|}{|\mathbf{v}|} = \frac{9}{3} = 3 \quad (219)$$

- (b) Construct the component equation of the plane that contains L_1 and the line L_2 . The line L_2 has the parametric equations $x = 2 + 2t$, $y = 1$, $z = -3t$.

Solutions: Direction vectors for the lines are:

$$L_1 : \quad \mathbf{v} = \langle 1, -2, 2 \rangle \quad (220)$$

$$L_2 : \quad \mathbf{u} = \langle 2, 0, 3 \rangle \quad (221)$$

The plane contains the direction vectors of both lines, and a vector perpendicular to them is

$$\mathbf{n} = \mathbf{v} \times \mathbf{u} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -2 & 2 \\ 2 & 0 & -3 \end{vmatrix} = \langle 6 - 0, -(-3 - 4), 0 - 4 \rangle = \langle 6, 7, 4 \rangle \quad (222)$$

A point in both lines is $P(2, 1, 0)$. The plane has component equation

$$6(x - 2) + 7(y - 1) + 4z = 0 \quad (223)$$

A few notes about the solution to part (b).

- It is not necessary to simplify further. But we could, for example, write the equation of the plane as

$$6x + 7y + 4z = 19$$

- Note that we can use other points instead of P , but the point that is used must be in the plane.
- Likewise we can use any two vectors that are parallel to the plane and are not parallel to each other. So instead of using $\mathbf{u}$ and $\mathbf{v}$, we can use other vectors. But regardless of which vectors we use, their cross product would need to be some multiple of $\mathbf{n} = \langle 6, 7, 4 \rangle$.

6. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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7. (5 points)

An object is moving along a path given by $\mathbf{r}(t)$ for $t \geq 0$, where $\mathbf{r}'(t) = \mathbf{v} = \langle t, 1, \sqrt{2t} \rangle$. Please show your work for the parts below. You can use your results for each part using results from previous parts of this question.

- (a) Determine the speed and the acceleration vector for the curve.

$$s = |\mathbf{v}| = \sqrt{t^2 + 1^2 + (\sqrt{2t})^2} = \sqrt{t^2 + 2t + 1} = \sqrt{(t+1)^2} = t+1 \quad (224)$$

$$\frac{d}{dt} \sqrt{2t} = \frac{1}{2} (2t)^{-1/2} \frac{d}{dt} (2t) = \frac{1}{\sqrt{2t}} \quad (225)$$

$$\mathbf{a} = \mathbf{v}'(t) = \langle 1, 0, \frac{1}{\sqrt{2t}} \rangle = \mathbf{i} + \frac{1}{\sqrt{2t}} \mathbf{k} \quad (226)$$

We can also express the acceleration vector in other ways. For example it ok to write it as

$$\mathbf{a} = \langle 1, 0, \frac{\sqrt{2}}{2\sqrt{t}} \rangle$$

- (b) Determine the length of the curve, L , from the point where $t = 2$ to $t = 8$. Simplify your answer as much as possible.

Apply the arc length formula.

$$L = \int_2^8 |\mathbf{v}| dt = \int_2^8 t+1 dt = \left(\frac{t^2}{2} + t \right) \Big|_2^8 = (32+8) - (2+2) = 36 \quad (227)$$

The question asked to simplify your work as much as possible, so for full credit you would **not** want to leave your answer as something like $40 - 4$.

- (c) Suppose the object starts at the origin, so that $\mathbf{r}(0) = \mathbf{0}$. Determine the value of t that corresponds to the point on the curve that is a distance of 6 units away from the origin along the curve. In other words, determine the value of t so that the arc length between the origin and $\mathbf{r}(t)$ is 6. Please show your work.

We can use the arc length parameter $s(t) = \int_{t_0}^t |\mathbf{v}| d\tau$, with $s = 10$, $t_0 = 0$, and solve for t .

$$s(t) = \int_{t_0}^t |\mathbf{v}| d\tau = 6 = \int_0^t \tau + 1 d\tau \quad (228)$$

$$6 = \left(\frac{\tau^2}{2} + \tau \right) \Big|_0^t \quad (229)$$

$$6 = \frac{t^2}{2} + t \quad (230)$$

$$0 = t^2 + 2t - 12 \quad (231)$$

$$t = -1 \pm \frac{1}{2} \sqrt{2^2 - 4 \cdot (-12)} = -1 \pm \frac{1}{2} \sqrt{52} \quad (232)$$

We should ignore the negative time because we were given that $t \geq 0$. So the value of t is $t = -1 + \sqrt{52}/2$. It is ok to simplify further, for example

$$t = -1 + \sqrt{52}/2 = -1 + \sqrt{13 \cdot 4}/2 = -1 + \sqrt{13}$$

Solutions

3.8 Year 1 MATH 2551 Exam 1 Sample H

Work done on scratch paper will not be graded. Please justify your reasoning on all questions unless indicated otherwise. Good luck on the exam!

A Few Helpful Formulas

$$d = \left| \mathbf{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|, \quad d = \frac{|\mathbf{PS} \times \mathbf{v}|}{|\mathbf{v}|}, \quad \mathbf{N} = \frac{d\mathbf{T}'(t)/dt}{|d\mathbf{T}/dt|}, \quad \kappa = \frac{|\mathbf{T}'(t)|}{|\mathbf{v}|} = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}, \quad \kappa = \frac{|f''(x)|}{[1 + (f'(x))^2]^{3/2}}$$

- (0.5 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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, the High School you attend: .
- (7 points) Fill in the blanks. You do not need to justify your reasoning.

- (a) The cosine of the angle between the vectors $\langle 2, 3, -6 \rangle$ and $\langle 2, 2, 1 \rangle$ is $\cos \theta =$.

Answer: 4/21

Solutions: $\cos \theta = \frac{\langle 2, 3, -6 \rangle \cdot \langle 2, 2, 1 \rangle}{\sqrt{2^2 + 3^2 + 6^2} \sqrt{2^2 + 2^2 + 1}} = \frac{4}{7 \cdot 3} = \frac{4}{21}.$

- (b) The volume of the parallelepiped determined by $\langle 1, 0, 1 \rangle$, $\langle 1, 1, 0 \rangle$, $\langle 0, 0, 2 \rangle$ is .

Solutions: Using the triple scalar product,

$$V = |\langle 1, 0, 1 \rangle \cdot (\langle 1, 1, 0 \rangle \times \langle 0, 0, 2 \rangle)|, \text{ and } \langle 1, 1, 0 \rangle \times \langle 0, 0, 2 \rangle = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{vmatrix} = \langle 2, -2, 0 \rangle$$

$$\begin{aligned} V &= |\langle 1, 0, 1 \rangle \cdot \langle 2, -2, 0 \rangle| \\ &= |2| \\ &= 2 \end{aligned}$$

- (c) The plane that passes through $P(0, -2, 2)$ and contains the line $x = 1 + t$, $y = 2t$, $z = 2 - t$ is . The distance between P and the xz -plane is .

Solutions: The plane is parallel to direction vector of given line, $\mathbf{v} = \langle 1, 2, -1 \rangle$. Line contains $Q(1, 0, 2)$, so plane also parallel to the vector $\mathbf{PQ} = \langle 1, 0, 2 \rangle - \langle 0, -2, 2 \rangle = \langle 1, 2, 0 \rangle$. It would also be ok to use any scalar multiple of this.

A normal to plane is

$$\mathbf{n} = \mathbf{v} \times \mathbf{PQ} = \begin{vmatrix} i & j & k \\ 1 & 2 & -1 \\ 1 & 2 & 0 \end{vmatrix} = \langle 2, -1, 0 \rangle$$

So using point $P(0, -2, 2)$ and $\mathbf{n}$, the plane has equation

$$2(x - 0) - (y + 2) + (0)(z - 2) = 0 \quad (233)$$

which could be simplified to $2x - y = 2$, but it isn't necessary to simplify the equation. The point S is 2 units away from the yz -plane because the point has coordinates $(0, -2, 2)$. So the distance between S and the yz -plane is 2.

- (d) Identify the surface $-x^2 - y^2 + z = 0$ by type (cone, ellipsoid, etc.).

Answer: elliptical paraboloid.

- (e) The position of a moving object is given by the curve $\mathbf{r}(t) = \langle 1 - 2t, 6t^3 \rangle$ for $t \in \mathbb{R}$. The curve passes through the point $P(5, -48)$ when $t = \text{$. The times when the direction of motion is perpendicular to $\mathbf{w} = \langle 1, 1 \rangle$, are $t = \text{$ and $t = \text{$.

Solutions: The point will be at P when

$$\mathbf{r}(t) = \langle 1 - 2t, 6t^3 \rangle = \langle 5, -48 \rangle \quad (234)$$

Solving for t we see that $t = -2$. The direction of motion is parallel to the velocity.

$$\mathbf{r}'(t) = \mathbf{v} = \langle -2, 18t^2 \rangle \quad (235)$$

Vectors are perpendicular when their dot products are zero. So the velocity will be orthogonal to the given $\mathbf{w}$ vector when

$$0 = \mathbf{v} \cdot \mathbf{w} = \begin{pmatrix} -2 \\ 18t^2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = -2 + 18t^2 \quad (236)$$

$$= 9t^2 - 1 \quad (237)$$

$$t = \pm 1/3 \quad (238)$$

- (f) If the velocity of an object at time t is $\mathbf{v}(t) = \langle 4t^3, \cos(t), 4e^{2t} \rangle$ and its position at $t = 0$ is $\mathbf{r} = \langle 3, 5, 14 \rangle$, then $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, where $f(t) = \text{$, $g(t) = \text{$, $h(t) = \text{$.

Solutions: Integrating $\mathbf{r}(t)$:

$$\mathbf{r} = \int \mathbf{v}(t) dt = \langle t^4 + c_1, \sin(t) + c_2, 2e^{2t} + c_3 \rangle \quad (239)$$

And $\mathbf{r}(0) = \langle c_1, c_2, 2 + c_3 \rangle = \langle 3, 5, 14 \rangle$. Therefore, $\mathbf{r} = \langle t^4 + 3, \sin(t) + 5, 2e^{2t} + 12 \rangle$.

- (g) A plane curve $\mathbf{r}(t)$ has curvature $\kappa = \frac{1}{4}$ at $\mathbf{r}(t_0) = \langle 5, 3 \rangle$, and the unit normal vector at $t = t_0$ is $\mathbf{N} = \mathbf{j} = \langle 0, 1 \rangle$. The equation of the osculating circle at $t = t_0$ is .

Answer: Solutions: Recall that The circle of curvature, or osculating circle, at a point P on a plane curve where $\kappa \neq 0$ is the circle in the plane of the curve that

- is tangent to the curve at P (has the same tangent line the curve has)
- has the same curvature the curve has at P

iii. lies toward the concave or inner side of the curve

The **radius of curvature** of the curve at P is the radius of the circle of curvature, which is $\frac{1}{\kappa}$. So to obtain the radius, we calculate κ and take the reciprocal. The **center** of the osculating circle of the curve at P lies on the inner side of the curve, and the unit normal points in the direction of the inner side of the planar curve.

So for this problem we have a circle with radius $1/\kappa = 4$, and whose centre is 4 units away from the point $(5, 3)$ in the direction of $\mathbf{N}$. The circle has equation

$$(x - 5)^2 + (y - 7)^2 = 4^2$$

3. (2 points) Indicate **true** if the statement is true, indicate **false** otherwise. You do not need to justify your reasoning.

true false

-
- | | |
|--|---|
| a) The set of points equidistant from a coordinate plane and a point off of that plane in $\mathbb{R}^3$ forms an elliptic paraboloid. Solution: This is true. For example, taking the xy -plane and the point $(0, 0, 1)$. The distance of the point (x, y, z) to the plane is clearly $ z $ and to our chosen point is $\sqrt{x^2 + y^2 + (z - 1)^2}$. Equating these and squaring yields $z^2 = x^2 + y^2 + (z - 1)^2$ which simplifies to $z = \frac{x^2}{2} + \frac{y^2}{2} + \frac{1}{2}$. This is the formula for the elliptic paraboloid and other choices of plane and point will simply yield other rotated and translated elliptic paraboloids. | ☐ ☐ |
| b) The surface $-3x^2 + 5y^2 - z^2 = 1$ is a cone along the z -axis. Solution: False. It is a hyperboloid of two sheets that opens up over the y -axis. | ☐ ☐ |
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4. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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5. (5 points) Suppose R is the point $R(0, 1, 2)$, L is the line $\mathbf{r}(t) = \langle 3t, 2, 4t \rangle$. Please show your work.

- (a) Construct the equation of the plane that passes through R and contains the line L .

Solutions: Two points on the line are $P(0, 2, 0)$ and $Q(3, 2, 4)$. So two vectors parallel to the plane are $\mathbf{PR}$ and $\mathbf{QR}$, where

$$\mathbf{PR} = R - P = \langle 0, 1, 2 \rangle - \langle 0, 2, 0 \rangle = \langle 0, -1, 2 \rangle$$

$$\mathbf{QR} = R - Q = \langle 0, 1, 2 \rangle - \langle 3, 2, 4 \rangle = \langle -3, -1, -2 \rangle$$

A normal vector to the plane is

$$\mathbf{PR} \times \mathbf{QR} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -1 & 2 \\ -3 & -1 & -2 \end{vmatrix} = \langle 4, -6, -3 \rangle$$

The plane has equation

$$4(x - 0) + (-6)(y - 1) + (-3)(z - 2) = 0$$

Note: The above is sufficient but we could also rearrange this equation to obtain, for example,

$$4x - 6y - 3z + 12 = 0$$

- (b) Calculate the distance between the point R and the line L .

Solutions: A point on the line that corresponds to where $t = 0$ is $P(0, 2, 0)$. The line has direction vector

$$\mathbf{v} = \langle 3, 0, 4 \rangle$$

The distance d can be obtained with the formula

$$d = \frac{|\mathbf{PR} \times \mathbf{v}|}{|\mathbf{v}|} \quad (240)$$

And

$$\mathbf{PR} = R - P = \langle 0, 1, 2 \rangle - \langle 0, 2, 0 \rangle = \langle 0, -1, 2 \rangle \quad (241)$$

$$\mathbf{PR} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -1 & 2 \\ 3 & 0 & 4 \end{vmatrix} = \langle -4, 6, 3 \rangle \quad (242)$$

$$|\mathbf{PR} \times \mathbf{v}| = \sqrt{4^2 + 6^2 + 3^2} = \sqrt{16 + 36 + 9} = \sqrt{61} \quad (243)$$

$$|\mathbf{v}| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5 \quad (244)$$

$$d = \frac{|\mathbf{PR} \times \mathbf{v}|}{|\mathbf{v}|} = \sqrt{61}/5 \quad (245)$$

Unsimplified answers are acceptable. It is ok to leave the answer as $\sqrt{61}/5$.

6. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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7. (5 points)

Consider the curve $\mathbf{r}(t) = 6 \cos t \mathbf{i} + 6 \sin t \mathbf{j} + 8t \mathbf{k}$, $t \in \mathbb{R}$. Please show your work for the following.

- a) Determine the unit tangent vector for this curve.

Solutions: the unit tangent vector is determined by the formula

$$\mathbf{T} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$$

Computing the necessary quantities:

$$\mathbf{v} = \mathbf{r}' = -6 \sin t \mathbf{i} + 6 \cos t \mathbf{j} + 8 \mathbf{k} \quad (246)$$

$$\|\mathbf{v}\|^2 = (-6 \sin t)^2 + (6 \cos t)^2 + (8)^2 \quad (247)$$

$$= 36 \sin^2 t + 36 \cos^2 t + 64 \quad (248)$$

$$= 36 + 64 \quad (249)$$

$$= 100 \quad (250)$$

$$\|\mathbf{v}\| = \sqrt{100} = 10 \quad (251)$$

$$\mathbf{T} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = -\frac{6}{10} \sin t \mathbf{i} + \frac{6}{10} \cos t \mathbf{j} + \frac{8}{10} \mathbf{k} \quad (252)$$

- b) Determine the arc length parameter $s(t)$ along the curve from the point where $t_0 \in \mathbb{R}$ by evaluating the integral

$$s = s(t) = \int_{t_0}^t \left| \frac{d\mathbf{r}}{d\tau} \right| d\tau = \int_{t_0}^t |\mathbf{v}| d\tau$$

Solutions: using the given formula we obtain

$$s(t) = \int_{t_0}^t |\mathbf{v}| d\tau = \int_{t_0}^t 10 d\tau = 10(t - t_0) \quad (253)$$

- c) The point $P_0(6, 0, 16\pi)$ is a point on the curve that corresponds to $t_0 = 2\pi$. Use your results from part (b) to determine the coordinates of any point on the curve that is a distance of 10π units **along the curve** from P_0 . Give your answers in terms of a point in $\mathbb{R}^3$.

Solutions: We were given that

$$\mathbf{r}(t) = 6 \cos t \mathbf{i} + 6 \sin t \mathbf{j} + 8t \mathbf{k}, \quad t \in \mathbb{R}$$

We know that if the $\mathbf{k}$ component of P_0 is 16π then $t = 2\pi$. Then we can use the arc length parameterization formula to obtain the values of t that give us the needed distance.

$$\text{point above } P_0 : \int_{2\pi}^{t_0} 10 \, d\tau = 10(t_0 - 2\pi) = 10\pi \Rightarrow t_0 = 3\pi \quad (254)$$

$$\text{point below } P_0 : \int_{t_0}^{2\pi} 10 \, d\tau = 10(-t_0 + 2\pi) = 10\pi \quad (255)$$

$$20\pi - 10t_0 = 10\pi \Rightarrow t_0 = \pi \quad (256)$$

The point above P_0 corresponds to $t = 3\pi$, and the point below corresponds to $t = \pi$. The two points are

$$t = \pi : \mathbf{r}(\pi) = -6\mathbf{i} + 0\mathbf{j} + 8\pi\mathbf{k}, \quad (-6, 0, 8\pi) \quad (257)$$

$$t = 3\pi : \mathbf{r}(3\pi) = -6\mathbf{i} + 24\pi\mathbf{k}, \quad (-6, 0, 24\pi) \quad (258)$$

We only needed to find one point, so either point is sufficient.

Solutions

3.9 Year 1 MATH 2551 Exam 1 Sample I

Work done on scratch paper will not be graded. Please justify your reasoning on all questions unless indicated otherwise. Good luck on the exam!

A Few Helpful Formulas

$$d = \left| \mathbf{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|, \quad d = \frac{|\mathbf{PS} \times \mathbf{v}|}{|\mathbf{v}|}, \quad \mathbf{N} = \frac{d\mathbf{T}'(t)/dt}{|d\mathbf{T}/dt|}, \quad \kappa = \frac{|\mathbf{T}'(t)|}{|\mathbf{v}|} = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}, \quad \kappa = \frac{|f''(x)|}{[1 + (f'(x))^2]^{3/2}}$$

1. (0.5 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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, the High School you attend: .
2. (7 points) Fill in the blanks. You do not need to justify your reasoning.

- (a) The midpoint of the line segment that joins points $P(1, 5, 2)$ and $Q(9, 3, 2)$ is $S(a, b, c)$, where $a = \boxed{}$, $b = \boxed{}$, $c = \boxed{}$.

Solutions: The midpoint is found by taking the average of the corresponding coordinates of the two points. The midpoint will be the point $(5, 4, 2)$. The midpoint of a line was covered in Section 12.2.

- (b) The parallelogram determined by vectors $\mathbf{a}$ and $\mathbf{b}$ has area $4\sqrt{2}$. If $\mathbf{a} = \mathbf{b} + \mathbf{c}$, and $\mathbf{c}$ makes an angle of $\frac{\pi}{4}$ with $\mathbf{b}$, and $|\mathbf{c}| = 2$, then $|\mathbf{b}| = \boxed{}$.

Solutions:

$$\begin{aligned}
 4\sqrt{2} &= |\mathbf{a} \times \mathbf{b}| \\
 &= |(\mathbf{b} + \mathbf{c}) \times \mathbf{b}| \\
 &= |\mathbf{b} \times \mathbf{b} + \mathbf{c} \times \mathbf{b}| \\
 &= |\mathbf{c} \times \mathbf{b}| \\
 &= |\mathbf{c}| |\mathbf{b}| \sin \theta \\
 &= 2 |\mathbf{b}| \frac{1}{\sqrt{2}} \\
 4(\sqrt{2})^2 &= 2|\mathbf{b}| \\
 |\mathbf{b}| &= 4
 \end{aligned}$$

- (c) The cosine of the acute angle, θ , between planes $3x + 4z = 1$ and $4x + 3y = 2$ is $\cos \theta = \boxed{}$.

Answer: 12/25

Solutions: the normals to the two planes can be:

$$\mathbf{n}_1 = \langle 3, 0, 4 \rangle \quad (259)$$

$$\mathbf{n}_2 = \langle 4, 3, 0 \rangle \quad (260)$$

Thus,

$$\cos \theta = \frac{\langle 3, 0, 4 \rangle \cdot \langle 4, 3, 0 \rangle}{\sqrt{3^2 + 4^2} \sqrt{3^2 + 4^2}} = \frac{12}{25}$$

Note that we could use other vectors for the normals. But whatever vectors we choose, the question requires that the angle be acute. So $0 \leq \theta \leq \pi/2$, which means $\cos \theta \geq 0$.

- (d) Identify the surface $-x^2 + y - 2z^2 = 0$ by type (cone, ellipsoid, etc.). .

Answer: paraboloid, or elliptical paraboloid.

- (e) Suppose that the tangent line to the curve $\mathbf{r}(t) = \langle 2t - 1, t - 3, 2t^2 \rangle$ at the point where $t = t_0 = 3$ is $\mathbf{L} = \langle f(u), g(u), h(u) \rangle$, $u \in \mathbb{R}$. Then the parametric equations for the tangent line at t_0 , with parameter u , are $f(u) = \text{}$, $g(u) = \text{}$, $h(u) = \text{}$. The curve intersects the plane $x + 2y = 4$ when $t = \text{}$.

Solutions: The tangent line is parallel to $\mathbf{r}'(t)$ and passes through $\mathbf{r}(3)$.

$$\mathbf{L}(u) = \mathbf{r}(3) + \mathbf{r}'(3)u \quad (261)$$

We need:

$$\mathbf{r}(3) = \langle 5, 0, 18 \rangle \quad (262)$$

$$\mathbf{r}'(t) = \langle 2, 1, 4t \rangle \quad (263)$$

$$\mathbf{r}'(3) = \langle 2, 1, 12 \rangle \quad (264)$$

Thus

$$\mathbf{L}(u) = \mathbf{r}(3) + \mathbf{r}'(3)u \quad (265)$$

$$= \langle 5, 0, 18 \rangle + \langle 2u, u, 12u \rangle \quad (266)$$

$$= \langle 5 + 2u, u, 18 + 12u \rangle \quad (267)$$

So

$$f(u) = 5 + 2u \quad (268)$$

$$g(u) = u \quad (269)$$

$$h(u) = 18 + 12u \quad (270)$$

The curve intersects the plane when the components of the curve satisfy the plane equation.

$$4 = x + y + z \quad (271)$$

$$= 2t - 1 + 2(t - 3) \quad (272)$$

$$= 4t - 7 \quad (273)$$

$$7 = 4t \quad (274)$$

The point corresponds to $t = 7/4$.

- (f) Consider the initial value problem (IVP): $\mathbf{r}' = 32t\mathbf{i} + 6t^2\mathbf{j} + 4e^{2t}\mathbf{k}$, $\mathbf{r}(0) = 3\mathbf{i} + 5\mathbf{j} + 8\mathbf{k}$. The solution to the IVP is $\mathbf{r} = \langle f(t), g(t), h(t) \rangle$, where $f(t) = \boxed{}$, $g(t) = \boxed{}$, and $h(t) = \boxed{}$.

Solutions:

$$f(t) = \int 32t \, dt = 16t^2 + c_1 \quad (275)$$

$$g(t) = \int 6t^2 \, dt = 2t^3 + c_2 \quad (276)$$

$$h(t) = \int 4e^{2t} \, dt = 2e^{2t} + c_3 \quad (277)$$

Then using $\mathbf{r}(0)$, we find $c_1 = 3$, $c_2 = 5$, $c_3 = 6$. So

$$f(t) = 16t^2 + 3 \quad (278)$$

$$g(t) = 2t^3 + 5 \quad (279)$$

$$h(t) = 2e^{2t} + 6 \quad (280)$$

Don't forget that $e^0 = 1$, so $h(0) = 2e^0 + c_3 = 8$ implies $c_3 = 6$.

- (g) The velocity vector of an object moving on the curve $\mathbf{r}(t)$ is $\mathbf{v}(t) = \langle t, 3 \rangle$ for $t > 0$. The speed of the object at $t = 2$ is $s(2) = |\mathbf{v}(2)| = \boxed{}$. The unit tangent vector at $t = 2$ is $\mathbf{T}(2) = \langle c_1, c_2 \rangle$, where $c_1 = \boxed{}$, $c_2 = \boxed{}$. The curvature is $\boxed{}$.

Answer: Solutions: The speed is the magnitude of the given velocity vector and is

$$s = |\mathbf{v}| = \sqrt{(t)^2 + (3)^2} = \sqrt{t^2 + 9} \quad (281)$$

$$s(2) = \sqrt{2^2 + 9} = \sqrt{13} \quad (282)$$

We also need the unit tangent vector at $t = 2$.

$$\mathbf{T}(t) = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{t\mathbf{i} + 3\mathbf{j}}{\sqrt{t^2 + 9}} \quad (283)$$

$$\mathbf{T}(2) = \frac{2\mathbf{i} + 3\mathbf{j}}{\sqrt{2^2 + 9}} = \left\langle \frac{2}{\sqrt{13}}, \frac{3}{\sqrt{13}} \right\rangle \quad (284)$$

$$c_1 = \frac{2}{\sqrt{13}} \quad (285)$$

$$c_2 = \frac{3}{\sqrt{13}} \quad (286)$$

The curvature can be found using the formula

$$\kappa = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3} = \frac{|\mathbf{v} \times \mathbf{a}|}{s^3} \quad (287)$$

Then expressing the vectors as quantities in $\mathbb{R}^3$ we obtain:

$$\mathbf{v} = \langle t, 3, 0 \rangle \quad (288)$$

$$\mathbf{a} = \langle 1, 0, 0 \rangle \quad (289)$$

$$|\mathbf{v} \times \mathbf{a}| = |\langle 0, 0, -3 \rangle| = 3 \quad (290)$$

$$|\mathbf{v}|^3 = s^3 = \left(\sqrt{t^2 + 3^2} \right)^3 \quad (291)$$

$$\kappa = \frac{3}{\left(\sqrt{t^2 + 3^2} \right)^3} \quad (292)$$

3. (2 points) Indicate **true** if the statement is true, indicate **false** otherwise. You do not need to justify your reasoning.

true false

- a) If two intersecting planes are defined by the equations $A_1(x - x_1) + B_1(y - y_1) + C_1(z - z_1) = 0$ and $A_2(x - x_2) + B_2(y - y_2) + C_2(z - z_3) = 0$, then the cosine of the angle between them is given by $\cos \theta = \left| \frac{\langle A_1, B_1, C_1 \rangle}{\|\langle A_1, B_1, C_1 \rangle\|} \cdot \frac{\langle A_2, B_2, C_2 \rangle}{\|\langle A_2, B_2, C_2 \rangle\|} \right|$. **Solution:** This is true. Recall that alternate formulas for these planes are given by $\vec{n}_1 \cdot \vec{P}_1 P = 0$ and $\vec{n}_2 \cdot \vec{P}_2 P = 0$ where $\vec{n}_1 = \langle A_1, B_1, C_1 \rangle$ and $\vec{n}_2 = \langle A_2, B_2, C_2 \rangle$ are the normal vectors to these respective planes and $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ are the respective base points these planes pass through. Also, we defined the angle between two planes to be acute angle between their normal vectors. And with that note this is exactly the formula we had for the angle between two vectors applied to these normal vectors. First, they are normalized so their length doesn't affect the dot product. Second, the dot product is taken which we know comes out to the cosine of the angle between unit vectors. Thirdly, this dot product is absolute valued, because otherwise we won't always end up with the acute angle between the planes. This absolute value can be thought of negating the normal vectors chosen to find the pair which have the smallest angle between them as desired. Finally, we take the inverse cosine function to the actual angle back.
- Note, we define the angle between two planes without referring directly to vectors lying on the planes, because different choices between these vectors can lead to different answers. For example, consider that vectors on the line of intersection will always either be parallel or antiparallel regardless of the planes involved. In fact, to get the correct angle using vectors on the plane we have to ensure they are perpendicular to this line of intersection. Try to visually convince yourself why this is and see if you can formalize why this works.
- b) The surface $x^2 + y^2 - z^2 + 1 = 0$ is a hyperboloid of one sheet. **Solution:** False. The surface is not defined for $-1 \leq z \leq 1$, so this is a hyperboloid of **two** sheets, not one. Also, one way to approach these sorts of exercises is to look at what the surface looks like for constant x , y , or z . Usually setting one or two variables equal to a constant is sufficient but in the interest of completeness here is the process for all three:
- **Setting z to a constant:** for any fixed $z = k$, and $k^2 \geq 1$, the surface is $x^2 + y^2 = k^2 - 1$, which is a family of circles.
 - **Setting x to a constant:** for any fixed $x = k$, the surface is $y^2 - z^2 = -k^2 - 1$, which is a family of hyperbolas.
 - **Setting y to a constant:** for any fixed $y = k$, the surface is $x^2 - z^2 = -k^2 - 1$, which is a family of hyperbolas.

This is a hyperboloid of two sheets.

4. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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5. (5 points)

Consider the point $S(6, 5, 2)$ and the L_1 whose parametric equations are $x = 2 + t$, $y = 1 - 2t$, $z = 2t$. Please show your work for the following.

- (a) Calculate the distance between the point and line L_1 . *Solutions:* A point on the line that corresponds to where $t = 0$ is $P(2, 1, 0)$. The line has direction vector $\mathbf{v} = \langle 1, -2, 2 \rangle$. The distance d can be obtained with the formula

$$d = \frac{|\mathbf{PS} \times \mathbf{v}|}{|\mathbf{v}|} \quad (293)$$

And

$$\mathbf{PS} = \langle 6, 5, 2 \rangle - \langle 2, 1, 0 \rangle = \langle 4, 4, 2 \rangle \quad (294)$$

$$\mathbf{PS} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 4 & 2 \\ 1 & -2 & 2 \end{vmatrix} = \langle 12, -6, -12 \rangle \quad (295)$$

$$|\mathbf{PS} \times \mathbf{v}| = \sqrt{144 + 36 + 144} = \sqrt{324} = 18 \quad (296)$$

$$|\mathbf{v}| = \sqrt{1 + 4 + 4} = \sqrt{9} = 3 \quad (297)$$

$$d = \frac{|\mathbf{PS} \times \mathbf{v}|}{|\mathbf{v}|} = \frac{18}{3} = 6 \quad (298)$$

For full credit the cross product and the lengths should be calculated, but unsimplified answers are ok. For example it is ok to leave the answer as

$$\frac{\sqrt{324}}{\sqrt{9}}$$

- (b) Construct the component equation of the plane that contains L_1 and the line L_2 . The line L_2 has the parametric equations $x = 2 + 2t$, $y = 1$, $z = -3t$.

Solutions: Direction vectors for the lines are:

$$L_1 : \quad \mathbf{v} = \langle 1, -2, 2 \rangle \quad (299)$$

$$L_2 : \quad \mathbf{u} = \langle 2, 0, -3 \rangle \quad (300)$$

The plane contains the direction vectors of both lines, and a vector perpendicular to them is

$$\mathbf{n} = \mathbf{v} \times \mathbf{u} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -2 & 2 \\ 2 & 0 & -3 \end{vmatrix} = \langle 6, 7, 4 \rangle \quad (301)$$

A point in both lines is $P(2, 1, 0)$. The plane has component equation

$$6(x - 2) + 7(y - 1) + 4z = 0 \quad (302)$$

It is not necessary to simplify further.

6. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID: , the High School you attend: .

7. (5 points)

Consider the curve $\mathbf{r}(t) = 6 \cos t \mathbf{i} + 6 \sin t \mathbf{j} + 8t \mathbf{k}$, $t \in \mathbb{R}$. Please show your work for the following.

- a) Determine the unit tangent vector for this curve.

Solutions: the unit tangent vector is determined by the formula

$$\mathbf{T} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$$

Computing the necessary quantities:

$$\mathbf{v} = \mathbf{r}' = -6 \sin t \mathbf{i} + 6 \cos t \mathbf{j} + 8 \mathbf{k} \quad (303)$$

$$\|\mathbf{v}\|^2 = (-6 \sin t)^2 + (6 \cos t)^2 + (8)^2 \quad (304)$$

$$= 36 \sin^2 t + 36 \cos^2 t + 64 \quad (305)$$

$$= 36 + 64 \quad (306)$$

$$= 100 \quad (307)$$

$$\|\mathbf{v}\| = \sqrt{100} = 10 \quad (308)$$

$$\mathbf{T} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = -\frac{6}{10} \sin t \mathbf{i} + \frac{6}{10} \cos t \mathbf{j} + \frac{8}{10} \mathbf{k} = -\frac{3}{5} \sin t \mathbf{i} + \frac{3}{5} \cos t \mathbf{j} + \frac{4}{5} \mathbf{k} \quad (309)$$

- b) Determine the arc length parameter $s(t)$ along the curve from the point where $t_0 \in \mathbb{R}$ by evaluating the integral

$$s = s(t) = \int_{t_0}^t \left| \frac{d\mathbf{r}}{d\tau} \right| d\tau = \int_{t_0}^t |\mathbf{v}| d\tau$$

Solutions: using the given formula we obtain

$$s(t) = \int_{t_0}^t |\mathbf{v}| d\tau = \int_{t_0}^t 10 d\tau = 10(t - t_0) \quad (310)$$

- c) The point $P_0(6, 0, 16\pi)$ is a point on the curve that corresponds to $t_0 = 2\pi$. Use your results from part (b) to determine the coordinates of any point on the curve that is a distance of 10π units **along the curve** from P_0 . Give your answers in terms of a point in $\mathbb{R}^3$.

Solutions: We know that if the $\mathbf{k}$ component of P_0 is 48π then $t = 2\pi$. There are a few different ways to get the distance. For example we can use the arc length parameterization formula to obtain the values of t that give us the needed distance.

$$\text{point above } P_0 : \int_{2\pi}^{t_0} 10 d\tau = 10(t_0 - 2\pi) = 10\pi \Rightarrow t_0 = 3\pi \quad (311)$$

$$\text{point below } P_0 : \int_{t_0}^{2\pi} 10 d\tau = 10(-t_0 + 2\pi) = 10\pi \quad (312)$$

$$20\pi - 10t_0 = 10\pi \Rightarrow t_0 = \pi \quad (313)$$

The point above P_0 corresponds to $t = 3\pi$, and the point below corresponds to $t = \pi$. We are given

$$\mathbf{r}(t) = 6 \cos t \mathbf{i} + 6 \sin t \mathbf{j} + 8t \mathbf{k}$$

The two points are

$$t = \pi : \quad \mathbf{r}(\pi) = -6\mathbf{i} + 8\pi\mathbf{k}, \quad (-6, 0, 8\pi) \quad (314)$$

$$t = 3\pi : \quad \mathbf{r}(3\pi) = -6\mathbf{i} + 24\pi\mathbf{k}, \quad (-6, 0, 24\pi) \quad (315)$$

We only needed to find one point, so either point is sufficient.

Solutions

3.10 Year 1 MATH 2551 Exam 1 Sample J

Work done on scratch paper will not be graded. Please justify your reasoning on all questions unless indicated otherwise. Good luck on the exam!

A Few Helpful Formulas

$$d = \left| \mathbf{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|, \quad d = \frac{|\mathbf{PS} \times \mathbf{v}|}{|\mathbf{v}|}, \quad \mathbf{N} = \frac{d\mathbf{T}'(t)/dt}{|d\mathbf{T}/dt|}, \quad \kappa = \frac{|\mathbf{T}'(t)|}{|\mathbf{v}|} = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}, \quad \kappa = \frac{|f''(x)|}{[1 + (f'(x))^2]^{3/2}}$$

1. (0.5 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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, the High School you attend: .
2. (7 points) Fill in the blanks. You do not need to justify your reasoning.

- (a) The equation $x^2 + y^2 - 6y + z^2 + 4z = 12$ represents a sphere whose radius is $r =$. The center of the sphere is at the point $P(a, b, c)$ where $a =$, $b =$, $c =$.

Answer: Completing the square and rearranging like terms:

$$x^2 + y^2 - 6y + z^2 + 4z = 12 \quad (316)$$

$$x^2 + (y^2 - 6y + 9 - 9) + (z^2 + 4z + 4 - 4) = 12 \quad (317)$$

$$x^2 + (y - 3)^2 + (z + 2)^2 - 9 - 4 = 12 \quad (318)$$

$$x^2 + (y - 3)^2 + (z + 2)^2 = 25 \quad (319)$$

The sphere has radius 5 and is centered at the point $(0, 3, -2)$.

- (b) The parallelogram determined by vectors $\mathbf{a}$ and $\mathbf{b}$ has area 12. If $\mathbf{a} = \mathbf{b} + \mathbf{c}$, and $\mathbf{c}$ makes an angle of $\frac{\pi}{2}$ with $\mathbf{b}$, and $|\mathbf{c}| = 2$, then $|\mathbf{b}| =$.

Solutions:

$$\begin{aligned} 12 &= |\mathbf{a} \times \mathbf{b}| \\ &= |(\mathbf{b} + \mathbf{c}) \times \mathbf{b}| \\ &= |\mathbf{b} \times \mathbf{b} + \mathbf{c} \times \mathbf{b}| \\ &= |\mathbf{c} \times \mathbf{b}| \\ &= |\mathbf{c}| |\mathbf{b}| \sin \theta \\ &= 2 |\mathbf{b}| 1 \\ 12 &= 2 |\mathbf{b}| \\ |\mathbf{b}| &= 6 \end{aligned}$$

- (c) The plane that passes through $P(1, 3, 2)$ and contains the line $x = 1 + t$, $y = 2 + 3t$, $z = 2 + 2t$ is . The distance between the point P and the x -axis is .

Solutions: The plane is parallel to direction vector of given line, $\mathbf{v} = \langle 1, 3, 2 \rangle$. Line contains $Q(1, 2, 2)$, so the plane is also parallel to the vector $\mathbf{PQ} = \langle 1, 2, 2 \rangle - \langle 1, 3, 2 \rangle = \langle 0, -1, 0 \rangle$. It would also be ok to use any scalar multiple of this.

A normal to the plane is

$$\mathbf{n} = \mathbf{v} \times \mathbf{PQ} = \begin{vmatrix} i & j & k \\ 1 & 3 & 2 \\ 0 & -1 & 0 \end{vmatrix} = \langle 2, 0, -1 \rangle = 2\mathbf{i} - \mathbf{k}$$

So using point $P(1, 3, 2)$ and $\mathbf{n}$, the plane has equation

$$(2)(x - 1) + (0)(y - 3) + (-1)(z - 2) = 0 \quad (320)$$

which could be simplified to other forms, such as

$$2x - z = 0$$

But it isn't necessary to simplify the equation. The distance between the point and the x -axis is $\sqrt{3^2 + 2^2} = \sqrt{13}$.

- (d) Identify the surface $x - 5y^2 - 2z^2 = 0$ by type (cone, ellipsoid, etc.). .

Answer: paraboloid, or elliptical paraboloid.

- (e) The position of a moving object is given by the curve $\mathbf{r}(t) = \langle 1 - t, 4, 3t \rangle$ for $t \in \mathbb{R}$. The speed of the object is given by . A unit vector that gives the direction of motion at time t is $\mathbf{T} = \langle f(t), g(t), h(t) \rangle$, where $f = \text{}$, $g = \text{}$, $h = \text{}$.

Solutions: The speed is the magnitude of the velocity.

$$\mathbf{r}'(t) = \mathbf{v} = \langle -1, 0, 3 \rangle \quad (321)$$

$$|\mathbf{r}'(t)| = \sqrt{1 + 1 + (6t)^2} = \sqrt{1 + 0 + 3^2} = \sqrt{10} \quad (322)$$

We can write the speed as $\sqrt{10}$. We do not need to express the answer using any units. The unit tangent vector $\mathbf{T}$ gives the direction of motion at any time t .

$$\mathbf{T} = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} = \left\langle \frac{-1}{\sqrt{10}}, 0, \frac{3}{\sqrt{10}} \right\rangle \quad (323)$$

Therefore,

$$f = -1/\sqrt{10} \quad (324)$$

$$g = 0 \quad (325)$$

$$h = 3/\sqrt{10} \quad (326)$$

No further simplification needed.

- (f) Consider the initial value problem (IVP): $\mathbf{r}' = 32t\mathbf{i} + 6t^2\mathbf{j} + 4e^{2t}\mathbf{k}$, $\mathbf{r}(0) = 3\mathbf{i} + 5\mathbf{j} + 8\mathbf{k}$. The solution to the IVP is $\mathbf{r} = \langle f(t), g(t), h(t) \rangle$, where $f(t) = \boxed{}$, $g(t) = \boxed{}$, and $h(t) = \boxed{}$.

Solutions:

$$f(t) = \int 32t \, dt = 16t^2 + c_1 \quad (327)$$

$$g(t) = \int 6t^2 dt = 2t^3 + c_2 \quad (328)$$

$$h(t) = \int 4 dt = 2e^{2t} \quad (329)$$

Then using $\mathbf{r}(0)$, we find $c_1 = 3$, $c_2 = 5$, $c_3 = 8$. So

$$f(t) = 16t^2 + 3 \quad (330)$$

$$g(t) = 2t^3 + 5 \quad (331)$$

$$h(t) = 2e^{2t} + 6 \quad (332)$$

- (g) A plane curve $\mathbf{r}(t)$ has curvature $\kappa = \frac{1}{10}$ at $\mathbf{r}(t_0) = \langle 5, 0 \rangle$, and the unit normal vector at $t = t_0$ is $\mathbf{N} = \mathbf{j} = \langle 0, 1 \rangle$. The equation of the osculating circle at $t = t_0$ is .

Answer: Solutions: Recall that The circle of curvature, or osculating circle, at a point P on a plane curve where $\kappa \neq 0$ is the circle in the plane of the curve that

- i. is tangent to the curve at P (has the same tangent line the curve has)
- ii. has the same curvature the curve has at P
- iii. lies toward the concave or inner side of the curve

The **radius of curvature** of the curve at P is the radius of the circle of curvature, which is $\frac{1}{\kappa}$. So to obtain the radius, we calculate κ and take the reciprocal. The **center** of the osculating circle of the curve at P lies on the inner side of the curve, and the unit normal points in the direction of the inner side of the planar curve.

So for this problem we have a circle with radius $1/\kappa = 10$, and whose centre is 10 units away from the point $(5, 0)$ in the direction of $\mathbf{N}$. The circle has equation

$$(x - 5)^2 + (y - 10)^2 = 10^2$$

3. (2 points) Indicate **true** if the statement is true, indicate **false** otherwise. You do not need to justify your reasoning.

true false

-
- a) If $\mathbf{r}(t)$ is a differentiable vector function of t and $\|\mathbf{r}\|$ is constant, then $\mathbf{r} \cdot \mathbf{r}' = 0$. ☐ ☐

Solution: This is true. It is a theorem covered in the textbook and lectures. The idea behind it is that:

$$\|\mathbf{r}\|^2 = \text{constant} \quad (333)$$

$$\frac{d}{dt} \|\mathbf{r}\|^2 = 0 \quad (334)$$

$$\frac{d}{dt} (\mathbf{r} \cdot \mathbf{r}) = 0 \quad (335)$$

$$\mathbf{r}' \cdot \mathbf{r} + \mathbf{r} \cdot \mathbf{r}' = 0 \quad (336)$$

$$2\mathbf{r}' \cdot \mathbf{r} = 0 \quad (337)$$

Geometrically it means that when the length of $\mathbf{r}$ is constant then the position and velocity vectors are orthogonal.

- b) Two lines in $\mathbb{R}^3$ that are parallel to a plane are parallel to each other. **Solution:** ☐ ☐
 False. Consider the plane $z = 0$ and the lines $\mathbf{r}_1 = t\mathbf{i}$, and $\mathbf{r}_2 = t\mathbf{j}$. The two lines are parallel to the plane and intersect at a point. This exercise was one of the problems from the Webwork homework.
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4. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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5. (5 points)

Suppose R is the point $R(0, 1, 2)$, L is the line $\mathbf{r}(t) = \langle 3t, 2, 4t \rangle$. Please show your work.

- (a) Construct the equation of the plane that passes through R and contains the line L .

Solutions: Two points on the line are $P(0, 2, 0)$ and $Q(3, 2, 4)$. So two vectors parallel to the plane are $\mathbf{PR}$ and $\mathbf{QR}$, where

$$\mathbf{PR} = R - P = \langle 0, 1, 2 \rangle - \langle 0, 2, 0 \rangle = \langle 0, -1, 2 \rangle$$

$$\mathbf{QR} = R - Q = \langle 0, 1, 2 \rangle - \langle 3, 2, 4 \rangle = \langle -3, -1, -2 \rangle$$

A normal vector to the plane is

$$\mathbf{PR} \times \mathbf{QR} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -1 & 2 \\ -3 & -1 & -2 \end{vmatrix} = \langle 4, -6, -3 \rangle$$

The plane has equation

$$4(x - 0) + (-6)(y - 1) + (-3)(z - 2) = 0$$

Note: The above is sufficient but we could also rearrange this equation to obtain, for example,

$$4x - 6y - 3z + 12 = 0$$

- (b) Calculate the distance between the point R and the line L .

Solutions: The line has direction vector

$$\mathbf{v} = \langle 3, 0, 4 \rangle$$

The distance d can be obtained with the formula

$$d = \frac{|\mathbf{PR} \times \mathbf{v}|}{|\mathbf{v}|} \quad (338)$$

And

$$\mathbf{PR} = \langle 0, -1, 2 \rangle \quad (339)$$

$$\mathbf{PR} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -1 & 2 \\ 3 & 0 & 4 \end{vmatrix} = \langle -4, 6, 3 \rangle \quad (340)$$

$$|\mathbf{PR} \times \mathbf{v}| = \sqrt{16 + 36 + 9} = \sqrt{61} \quad (341)$$

$$|\mathbf{v}| = \sqrt{9 + 16} = \sqrt{25} \quad (342)$$

$$d = \frac{|\mathbf{PR} \times \mathbf{v}|}{|\mathbf{v}|} = \sqrt{61}/\sqrt{25} = \sqrt{61}/5 \quad (343)$$

Unsimplified answers are acceptable. It is ok to leave the answer in the form $\sqrt{61}/25$.

6. (0.25 points) Fill in the blanks with a dark pen or pencil. Using only capital letters print your first name: , last name: , the remaining digits of your GTID:

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7. (5 points)

An object is moving along a path given by $\mathbf{r}(t)$ for $t \geq 0$, where $\mathbf{r}'(t) = \mathbf{v} = \langle t, 1, \sqrt{2t} \rangle$. Please show your work for the parts below. You can use your results for each part using results from previous parts of this question.

- (a) Determine the speed and the acceleration vector for the curve.

$$s = |\mathbf{v}| = \sqrt{t^2 + 1^2 + (\sqrt{2t})^2} = \sqrt{t^2 + 2t + 1} = \sqrt{(t+1)^2} = t+1 \quad (344)$$

$$\frac{d}{dt} \sqrt{2t} = \frac{1}{2} (2t)^{-1/2} \frac{d}{dt} (2t) = \frac{1}{\sqrt{2t}} \quad (345)$$

$$\mathbf{a} = \mathbf{v}'(t) = \langle 1, 0, \frac{1}{\sqrt{2t}} \rangle = \mathbf{i} + \frac{1}{\sqrt{2t}} \mathbf{k} \quad (346)$$

- (b) Determine the length of the curve, L , from the point where $t = 2$ to $t = 8$. Simplify your answer as much as possible.

Apply the arc length formula.

$$L = \int_2^8 |\mathbf{v}| dt = \int_2^8 t+1 dt = \left(\frac{t^2}{2} + t \right) \Big|_2^8 = (32+8) - (2+2) = 36 \quad (347)$$

The question asked to simplify your work as much as possible, so for full credit you would **not** want to leave your answer as something like $40 - 4$.

- (c) Suppose the object starts at the origin, so that $\mathbf{r}(0) = \mathbf{0}$. Determine the value of t that corresponds to the point on the curve that is a distance of 6 units away from the origin along the curve. In other words, determine the value of t so that the arc length between the origin and $\mathbf{r}(t)$ is 6. Please show your work.

We can use the arc length parameter $s(t) = \int_{t_0}^t |\mathbf{v}| d\tau$, with $s = 10$, $t_0 = 0$, and solve for t .

$$s(t) = \int_{t_0}^t |\mathbf{v}| d\tau = 6 = \int_0^t \tau + 1 d\tau \quad (348)$$

$$6 = \left(\frac{\tau^2}{2} + \tau \right) \Big|_0^t \quad (349)$$

$$6 = \frac{t^2}{2} + t \quad (350)$$

$$0 = t^2 + 2t - 12 \quad (351)$$

$$t = -1 \pm \frac{1}{2} \sqrt{2^2 - 4 \cdot 1 \cdot (-12)} = -1 \pm \sqrt{13} \quad (352)$$

We should ignore $t = -1 - \sqrt{13}$ because we were given that $t \geq 0$. So the value of t is $t = -1 + \sqrt{13}$.